



Enhancing CCTA image quality: a review of deep learning approaches for advanced artifact correction and denoising

Mohanad Alkhodari¹ · Eman Alefisha² · Herbert F. Jelinek² ·
Ahmed Kaabneh³ · Panos Liatsis⁴

Accepted: 25 June 2025 / Published online: 5 August 2025
© The Author(s) 2025

Abstract

Cardiac imaging is vital for diagnosing coronary artery disease (CAD), with coronary computed tomography angiography (CCTA) being commonly used to evaluate coronary vessels for stenosis, calcification, and atherosclerosis. However, CCTA images often suffer from artifacts like beam hardening, scatter, and noise, degrading image quality and obscuring anatomical details, leading to diagnostic uncertainty. Conventional post-processing techniques, such as filtered back projection and iterative reconstruction, have limited effectiveness in correcting these artifacts, posing challenges in CCTA, where precise visualization of coronary arteries is crucial. Artifacts can blur vessel boundaries, obscure calcified plaques, and misrepresent stenosis severity, potentially leading to misdiagnosis and suboptimal clinical decisions. Recent advancements in computational imaging, particularly deep learning algorithms, offer clinical benefits for artifact reduction in CCTA. Deep learning models, such as convolutional neural networks (CNNs), outperform traditional methods by effectively de-noising and correcting artifacts through learning complex patterns from large datasets. These models adapt to the non-linear, heterogeneous nature of artifacts, enhancing image clarity and diagnostic reliability. Improved image quality in CCTA enables better visualization of coronary arteries, aiding in accurate assessment of stenosis and calcification. This review highlights deep learning approaches for artifact correction in CCTA, emphasizing their potential to improve CAD diagnosis.

Keywords Cardiac imaging · CCTA · CT imaging · De-noising · Photometric correction · Artifacts · Deep learning

✉ Panos Liatsis
panos.liatsis@ku.ac.ae

¹ Department of Biomedical Engineering & Biotechnology, Khalifa University, Abu Dhabi, United Arab Emirates

² Department of Medical Sciences, Khalifa University, Abu Dhabi, United Arab Emirates

³ Department of Interventional Radiology, NMC Royal Hospital, Abu Dhabi, United Arab Emirates

⁴ Department of Computer Science, Center for Secure Cyberphysical Systems, Khalifa University, Abu Dhabi, United Arab Emirates

1 Introduction

Coronary computed tomography angiography (CCTA) was first introduced in the early 2000 s when multi-detector row CT (MDCT) was developed to enhance current electron beam CT sensitivity to coronary calcium scoring (Schroeder et al. 2008). Three decades later, it has become a widely used non-invasive modality and the first-line test for coronary artery disease (CAD) assessment (Min et al. 2010). CAD accounts for nearly 6.7% of the adult population in the USA (around 18.2 million), making it the most common type of heart disease (Fryar et al. 2012). Thus, to provide proper assessment, CCTA is commonly used to understand the structural changes of the cardiac blood vessels in the presence of CAD owing to its high three-dimensional (3D) temporal, spatial, and volumetric resolutions for coronary arteries and vessels (Abdelrahman et al. 2020). CCTA can be utilized in applications that involve imaging atherosclerosis stages, plaque formation and progression, calcium scoring, and vessel geometry reconstruction (Moss et al. 2017). Due to its diverse applications, wide availability, simplicity, fast screening time, and high patient acceptance compared to magnetic resonance imaging (MRI), it is considered ideal in clinical settings (Infante et al. 2021).

Since the introduction of electron beam CT and the Agatston method for calcium deposition calculations (Sun et al. 2012), CT imaging, and more specifically, CCTA, has received attention with rapid technological developments. For instance, CCTA increased from the early 32-, 40-, and 64-slice scanners to dual-source CT, reaching a temporal resolution of 83 ms (Eckert et al. 2019). Moreover, the expansion to a 320-slice system allowed for capturing heart function even with high heart rates in one full gantry rotation (Li et al. 2020a). However, despite these major advances in the technology behind CCTA, it has several limitations, including the use of high radiation dose, reduced image quality in obese patients, poor outcomes in the presence of arrhythmias, and over-estimation of densely calcified vessels (Abdelrahman et al. 2020).

Taking CCTA limitations into consideration, as in most imaging modalities, the produced images are prone to noise contamination from multiple sources (Kay et al. 2020). For example, since the radiation dose can be harmful to the subject being imaged, a dose reduction is often recommended. However, this introduces noise in the generated images, which downgrades their quality and diagnostic value, particularly in applications related to fine structures, e.g., detection of small vessels (Abdullah et al. 2016). Moreover, it may suffer from other artifacts including the blooming effect of calcium deposits, beam hardening caused by metal objects, and other motion-based deformations (Andrew and John 2015).

Various solutions to handle such artifacts and the effects of noise without negatively affecting the quality and diagnostic value of CCTA images have been proposed (Sabarudin et al. 2013). Several studies have investigated the use of iterative algorithms, e.g., iterative Reconstruction (IR), demonstrated potential in reducing noise such as the occurrence of blooming artifacts compared to filtered back-projection (FBP) Jeon et al. (2023). Model-based iterative reconstruction (MBIR) methods that rely on repeated forward and backward projections have also been investigated (Benz et al. 2018). In presence of the modern dual-

source CT, virtual monoenergetic imaging (VMI) was used in image post-processing at different kiloelectron volt (keV) settings Grant et al. (2014), demonstrating improvements in the contrast-to-noise ratio in several applications, e.g., reducing blooming artifacts (Stehli et al. 2016) and handling high attenuation (Secchi et al. 2015); thus, improving the assessment of stenosis (Leithner et al. 2018). Several post-processing methods have also been applied to CT image denoising, including Wavelet transform (Wang et al. 2022b), dictionary learning (Zhang et al. 2015), and block-matching (Zhao et al. 2019). Specifically, a number of CCTA image restoration algorithms have been proposed, including Gaussian Mixture prior (GMP) Chen et al. (2015), locally-consistent nonlocal-means (LC-NLM) Green et al. (2016), and noise confidence region evaluation (NCRE) Hashemi et al. (2015). However, the majority of these methods have limitations that cause a loss in image quality thus, reducing diagnostic performance.

Recently, a number of deep learning applications emerged in the broad area of CT image post-processing and photometric correction (Liu et al. 2020b; Oostveen et al. 2021). Several studies reported enhancements in CT image generation, manifested through an apparent reduction in the amount of noise components when using deep learning reconstruction algorithms (Tatsugami et al. 2019; Benz et al. 2020; Catapano et al. 2023). Other studies used deep learning in the context of post-processing reconstructed images for the purposes of restoration and denoising (Li et al. 2020e, 2022; Kulathilake et al. 2023). The vast majority of these methods utilize convolution, encoding-decoding, or generative mechanisms; which ensure preserving the spatial hierarchies and extract local features (Sadia et al. 2024). Similar to any other deep learning model, de-noising models should be fine-tuned to overcome challenges, for instance, data issues (e.g., train dataset size, train vs test data distribution shift and image quality), potentially causing the models not to generalize well Kulathilake et al. (2023). To date, CT vendors incorporated deep learning methods within CT machines for image enhancement; examples include TrueFidelity (GE Healthcare), AiCE (Canon Medical Systems), and Precise Image (Philips Healthcare) Koetzier et al. (2023). Incorporating these deep learning models and CT image restoration into current equipment has improved diagnostic accuracy. Still, the potential of these models in CCTA image enhancement and applications for improving detection of stenosis, scoring calcified plaques, and providing proper coronary artery assessment has not been adequately addressed, necessitating a comprehensive review to explore current algorithms for improvement in CAD diagnostics with CCTA.

This review provides a thorough overview of the most recent deep learning-based applications for CCTA photometric image de-noising and artifact correction. We explore the causes of noise in CCTA, including their nature, effect, and mathematical models. We summarize the state-of-the-art studies that have used deep learning algorithms to perform post-processing restoration, correction, and image enhancement by reporting their contribution, performance, and limitations. Finally, we provide a thorough evaluation of these cutting-edge algorithms, assessing their current progress, while identifying inherent limitations and prevailing challenges.

2 Noise sources in CT imaging

2.1 Low-dose noise

2.1.1 Background

Low-dose CT imaging was produced first in the 1990 s Sone et al. (2001) to reduce the risks associated with regular CT scans. The radiation dose is typically reduced by a reduction in CT tube current (mAs), tube voltage (kVp), gantry speed rotation, and other pitch factor measures (Rampinelli et al. 2012). Reducing the X-ray tube's operating current generates low-dose CT scans and shortens the time of exposure to harmful beams (Kim et al. 2020). The dose of a CT scan can be calculated from the total number of photons being emitted (measured in milliamperere-seconds, mAs) Kovan (2024). An increase in the tube current results to an increase in the number of electrons, thus resulting in more X-rays being produced (Hermena and Young 2023).

The millisievert (mSv) measure is a unit of measurement used to quantify the dose of ionizing radiation absorbed by human tissues (Tootell et al. 2014). The average dose for regular CT imaging of the chest is 6.6 mSv, according to a large European study European Union (2014). In low-dose imaging, the dose is reduced to less than 1 mSv. A further decrease in the dose generates ultra-low-dose CTs reaching the level of plain X-ray Tækker et al. (2021).

2.1.2 Mathematical model

Noise may arise due to reduced transmitted radiation dose in CT imaging Mohammadinejad et al. (2021). In most cases, low-dose noise follows a Poisson distribution (Fig. 1), i.e., resembling the nature of additive noise, which gets affected by pixel values (Zhu et al. 2012; Yang et al. 2023). Poisson noise is a form of low light/radiation uncertainty measurement that is derived based on the signal itself rather than the detection process (Hasinoff 2021). During the transmission of photons, the sensors detect the number of discrete photon incidents over a given period, where this random per-photon arrival may cause noise similar to Poisson noise (Thanh and Surya 2019). Accordingly, the Poisson noise model ($f(x)$) is usually given as,

$$f(x) = \frac{e^{-\lambda T} (\lambda T)^x}{x!}, \quad (1)$$

where λ is the mean rate of transmitted photons in a given time interval (T), which is proportional to scene radiation, and x is a Poisson random variable that indicates the number of photons measured by a sensor. Evaluating this kind of noise requires knowledge about the standard deviation of the distribution (σ), which is equal to $\sqrt{\lambda T}$ Thanh and Surya (2019). Accordingly, the signal-to-noise (SNR) ratio can be calculated as,

$$SNR = \frac{\lambda T}{\sqrt{\lambda T}} \quad (2)$$

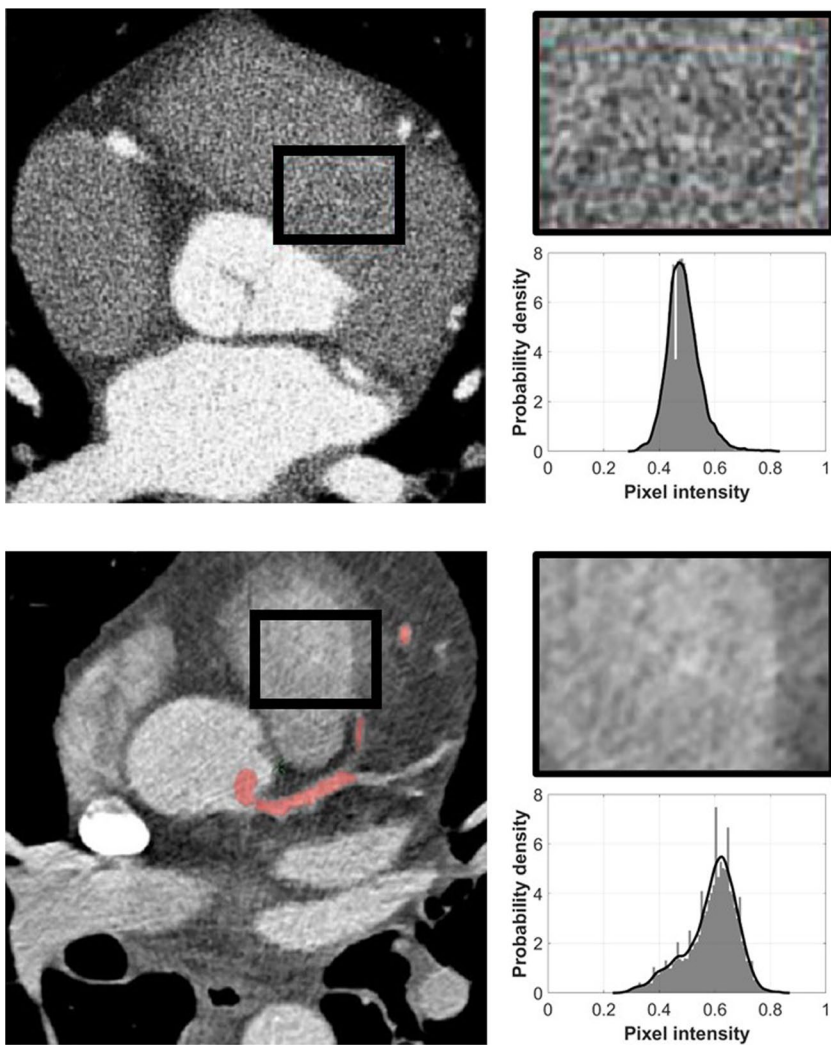


Fig. 1 Example of low-dose noise taken from coronary computed tomography angiography image. The example is for a CCTA slice with synthetically generated Poisson noise (upper) and a real noise low-dose CCTA slice (bottom). Retrieved from Green et al. (2018) with additional editing

2.1.3 Conventional approaches to de-noising

Clinicians prefer to reduce the exposure of X-ray beams to prevent risks associated with the ionization of tissues caused by these beams Goo (2012). However, this reduces the quality of CT images, often having a grainy appearance with unwanted changes in pixel values and image texture due to the high-frequency components (Padole et al. 2015; Gu et al. 2021). On the other hand, increasing the dose ensures higher-quality images with a clearer texture for tissues. However, this comes with a cost, and clinicians try to balance dose exposure and image contrast, resolution, and quality Grosser et al. (2019). Nowadays, CT machines

are embedded with techniques to enhance image quality and improve the overall diagnostic information, i.e., iterative reconstruction (IR) or filtered back-projection (FBP) algorithms (Solomon et al. 2017; Mileto et al. 2018, 2019). However, these techniques exhibit flaws in low-dose imaging. In particular, low-contrast lesions become blurred and may sometimes be undetectable due to the appearance of textural noise, i.e., blocky noise artefacts (Fujii et al. 2022). For instance, FBP is a linear mathematical operation, which is unable to effectively deal with the non-linear nature of CT images, causing changes to the spatial resolution (Nakaura et al. 2016).

2.1.4 Deep learning approaches to de-noising

Most recently, there has been wide interest on the use of deep learning in generating high quality images and its applications to de-noising (You et al. 2018; Immonen et al. 2022). The majority of these algorithms can be divided into discriminative, generative, or hybrid categories, each with distinct approaches to model learning and data representation (Kulathilake et al. 2023). The key modeling structure of discriminative algorithms relies on supervised deep convolutional neural networks (CNN) that learn from labeled data using densely connected layers (Chen et al. 2017; Kim et al. 2019; Huang et al. 2020a; Li et al. 2020b). Several studies proposed models that map between low-dose images and normal-dose high-quality images to learn the image restoration task with less noise, including dense networks (DenseNet) Ming et al. (2020), visual geometry group networks (VGG) Han et al. (2021), residual networks (ResNet) Yin et al. (2019), and stacked competitive networks (SCN) Du et al. (2017). These models have high performance in reducing artifacts and restoring low-dose images. However, they require the use of ground truth, dividing the images into smaller patches, applying several pre-processing steps, thus leading to over-smoothed structures due to the mean squared error (MSE)-based loss function Usui et al. (2021). Moreover, despite the high performance, most of these studies either utilized synthetic noise patterns or did not consider disease cases.

Alternatively, generative models tend to learn from the probabilistic distribution of the data and are primarily unsupervised, i.e., the requirement for annotated data is relaxed (Heinrich et al. 2018). For instance, auto-encoders have been proposed and found effective in predicting noiseless images from low-dose CT (Chen et al. 2017; Liu and Zhang 2018; Lee et al. 2018; Fan et al. 2019). However, despite their high signal-to-noise (SNR) ratio, the preservation of textural details and outline information is lacking. Furthermore, most studies have relied on MSE-based metrics when developing auto-encoder models Kulathilake et al. (2023). Another type of generative model that is used in low-dose CT image restoration is the U-Net network that has contracting and expanding paths (U-shaped) for learning image features (Feng et al. 2020; Chen et al. 2022; Jing et al. 2022). U-Net has an advantage over traditional auto-encoders, considering that it has concatenation skip connections, whereas the latter has summation connections Liu et al. (2020a). This enhances the learned information and improves performance when de-noising images.

Recent developments in deep learning allow for the implementation of hybrid models, i.e., using both generative and discriminative techniques in the form of generative adversarial networks (GAN) frameworks Vey et al. (2019). Here, the network learns from a generator that synthesizes samples from the low-dose images. In contrast, a discriminator network learns to distinguish between generated images and the original normal-dose

images. Several studies have focused on developing GAN of multiple structures including Wasserstein (W-GAN) Yin et al. (2019), Cycle (C-GAN) Kang et al. (2018), Tang et al. (2019), and Least squares (LS-GAN) Yang et al. (2019), Ma et al. (2020). These studies have shown that GAN can handle noise caused by low-dose imaging, while preserving fine details due to the utilization of multi-objective functions. Moreover, GAN can be developed as a combination of U-Nets and CNNs, which overcomes the drawbacks of both modeling techniques (Kulathilake et al. 2023). It is worth mentioning that the attention mechanism can be easily implemented within any of these deep learning models Li et al. (2020b), Li et al. (2021), Marcos et al. (2021), which adds another layer of learning during training. Here, the network tends to selectively emphasize on certain parts within the image that have important details during training, e.g., spatial features, channel maps, or global dependencies. Most recently, another category of generative approaches, i.e., diffusion models, have been introduced in several studies to investigate unsupervised learning without the need for noisy reference labels (Xia et al. 2022; Gao et al. 2023; Liu et al. 2023; Wang et al. 2024b). Owing to the functionality of diffusion models, low-dose noise patterns are learned through forward and reverse processes in a step-wise manner.

In the area of coronary CT angiography (CCTA), the regular dose ranges from 4 to 19 mSv Cousins et al. (2013), which is considered high if repeated scans are required, as is the case for lung scans and female breast scans (Nikolic et al. 2010). Therefore, low-dose imaging is often used in CCTA to overcome the risks above. However, as the main objective of CCTA is to screen coronary arteries and lumen to identify stenosis, calcified plaques, and calcium scores (Liu et al. 2020b), obtaining images with a clear texture and diagnostic quality is often difficult. Accordingly, CCTA studies utilizing deep learning techniques (Table 1) focused on de-noising low-dose images, while preserving the shape, texture, and outline of the entire coronary anatomy (Yang 2021). For example, discriminative three-dimensional (3D) and four-dimensional (4D) CNNs were used Green et al. (2018), Kobayashi et al. (2023) to apply 3D/4D convolutions, which enforced inter-slice continuity of tiny structures. Despite the high performance using CNNs with PSNR and SSIM reaching up to 47.00 dB and 0.938, respectively, current studies did not report performance in the presence of severe disease states, i.e., high calcification score. Generative U-Nets were shown to achieve high performance in several studies Ahn et al. (2019), Hong et al. (2020), Sandfort et al. (2024) with a high intra-class correlation coefficient with human experts, when segmenting the de-noised images Sandfort et al. (2024). Lastly, hybrid models, i.e., GANs, were utilized in various studies Gu et al. (2021), Sun and Ng (2022a), Deng et al. (2022) and have shown high performance in preserving the calcified plaques within the CCTA images especially those which suffer from blooming artifacts Sun and Ng (2022a). Compared with CT-based studies that tend to rely on simulated noise patterns, studies here utilized real data and reached PSNR levels up to 39.92 dB.

2.2 Blooming effect

In CCTA, accurate evaluation of coronary stenosis is essential, yet image distortions such as blooming artifacts pose significant challenges, particularly when assessing calcified lesions Ibrahim et al. (2017). In this scenario, the screened vessel appears to be larger than its actual size due to high-density structures, i.e., calcified plaques, causing false positives in CCTA diagnosis (Fig. 2). The blooming effect obscures the lumen, causing uncertainties in the

severity evaluation of the stenosis Andrew and John (2015). The clinically relevant size for coronary calcification is usually 1–3 mm in diameter, which allows for assessing the plaque clearness and width Pack et al. (2022). However, this characteristic gets deformed during blooming, causing poor diagnosis of stenosed arteries.

Another form of coronary blooming is caused by metal objects within the artery, i.e., stents, which usually occurs when evaluating in-stent restenosis (ISR) Yan et al. (2013), Pack et al. (2022). Due to metallic stents, the vessel at the location around the stent appears larger than its actual size in CCTA images, which causes luminal narrowing and a decrease in the intraluminal attenuation values Esposito et al. (2013). This issue is particularly detrimental during percutaneous coronary interventions (PCI), as it hinders accurate evaluation of stented segments (Wuest et al. 2013). Clinical guidelines recommend assessing stents with a minimum diameter of 3 mm to ensure reliable visualization of the lumen, thereby minimizing diagnostic errors during ISR evaluation and guiding appropriate clinical interventions (Members et al. 2010).

2.2.1 Mathematical model

The main cause for the blooming artifact appearance is sensor saturation, which in the case of CCTA occurs in the vicinity of calcified plaques (de Knecht et al. 2019). When applying a smoothing filter kernel, the intensities of image pixels around the area of interest exceed the dynamic range of the sensor and, thus, appear larger than usual with a brighter appearance (Kruk et al. 2014). This phenomenon can be usually caused due to partial volume averaging; which makes it harder to distinguish highly contrasted objects and delineate their edges (Barrett and Keat 2004). In contrast to high-frequency low-dose noise introduced in the previous section, the blooming artifact can be considered as a type of low-frequency blurry noise often formed because of the detector cell size, focal spot size, azimuthal blur, cross-talk, and reconstruction algorithm kernel (Sabarudin et al. 2013; Pack et al. 2022).

The mathematical model of the blooming artifact can be represented as a diffusion of excess charge from the saturated pixel to neighboring pixels as follows:

$$I(x, y) = \min(I(x, y), I_s), \quad (3)$$

where $I(x, y)$ is the intensity of the pixel (x, y) within the image and I_s is the saturation threshold.

The spread of the blooming artifact to neighboring pixels is based on a point spread function (PSF) Belloir et al. (2017), Triche et al. (2019) that generally illustrates the excess charge ($\Delta I(x, y)$) transmission:

$$\Delta I(x, y) = I(x, y) - I_s, \quad (4)$$

$$I_{bloom}(x, y) = I(x, y) + \sum_{(x', y') \in \Omega} PSF(x', y') \cdot \Delta I(x - x', y - y') \quad (5)$$

where Ω is the set of affected neighboring pixels and $PSF(x', y')$ represents the change in intensity in the neighboring pixels.

Table 1 Summary of the state-of-the-art studies on using deep learning for low-dose CT image restoration and de-noising

Study	Year	Modality	Organ	Image dimension	Dataset	Noise origin
Convolutional neural network (CNN)						
Green et al. (2018)	2018	CCTA	Heart	3D	45 scans (Local dataset)	Synthetic Poisson noise (ASTRA toolbox Van Aarle et al. 2016)
Li et al. (2020)	2020	CT	Abdomen Lungs	3D	10 patients – 2,378 images (CT Grand Challenge McCollough 2016)	Synthetic Poisson noise
Klug et al. (2022)	2022	CT	Heart	2D	60 patients – 60 scans (Local dataset)	Original noise
Jeon et al. (2023)	2022	CCTA	Heart	2D	90 patients – 90 scans (Local dataset)	Original noise
Kobayashi et al. (2023)	2023	CCTA	Heart	2D-4D	100 patients – 100 scans (Local dataset)	Original noise
Ortzi et al. (2023)	2023	CCTA	Heart	2D	70 patients – 70 scans (Local dataset)	Original noise
Generative adversarial network (GAN)						
Yang et al. (2019)	2019	CT	Abdomen Lungs	2D	10 patients – 2,378 images (CT Grand Challenge McCollough 2016)	Simulated noise
Kim et al. (2020)	2020	CT	Abdomen Lungs	2D	10 patients – 2,378 images (CT Grand Challenge McCollough 2016)	Simulated noise
Ma et al. (2020)	2020	CT	Lungs	2D	10 patients – 2,378 images (CT Grand Challenge McCollough 2016)	Simulated noise
Gu et al. (2021)	2021	CCTA	Heart	2D	149 patients – 149 scans (Local dataset)	Original noise
Usui et al. (2024)	2024	CT	Lungs	2D	40 patients – 4,250 images (Local dataset)	Synthetic noise (Sparse projections)

Table 1 (continued)

Study	Year	Modality	Organ	Image dimension	Dataset	Noise origin
U-Net						
Jing et al. (2022)	2022	CT	Abdomen Lungs	2D	10 patients – 2,378 images (CT Grand Challenge McCollough 2016)	Simulated noise
Sandfort et al. (2024)	2024	CCTA	Heart	3D	566 patients – 36,111 images (Local dataset)	Original noise
Diffusion models						
Xia et al. (2022)	2022	CT	Lungs	2D	10 patients – 5,936 images (CT Grand Challenge McCollough 2016)	Simulated noise
Gao et al. (2023)	2023	CT	Lungs	2D	10 patients – 5,936 images (CT Grand Challenge McCollough 2016)	Simulated noise
					299 scans - 800 images (Low-dose CT and Projection dataset Moen et al. 2021)	Synthetic Poisson noise
					850 images (Piglet dataset Yi and Babyn 2018)	
Liu et al. (2023)	2023	CT	Lungs	2D	10 patients (Low-dose CT and Projection dataset Moen et al. 2021)	Simulated noise

Table 1 (continued)

Study	Year	Modality	Organ	Image dimension	Dataset	Noise origin	
Wang et al. (2024)	2024	CT	Lungs	2D	10 patients – 5,936 images (CT Grand Challenge McCollough 2016)	Simulated noise	
Study	Training ground-truth	Network details	Input	Output	Noise/de-noising model	Performance	Limitations
Con-vol-utional neural net-work (CNN)	High-dose scans	3D convolutions	$15 \times 15 \times 3$ patches of noisy image	De-noised image	$x = \sigma(y)$	SSIM: 0.864 PSNR: 41.47	- Synthetic noise - Small dataset
	Green et al. (2018)						
Li et al. (2020)	Normal-dose scans	3D convolutions with self-attention blocks	$64 \times 64 \times 3$ patches of noisy image	De-noised image	$f(x) = \frac{e^{-\lambda T} (\lambda T)^x}{x!}$ $x = \sigma(y)$	SSIM: 0.888 PSNR: 27.74	- Small dataset - Slight deformation in the structure - No pixel-wise loss function
					$f(x) = \frac{e^{-\lambda T} (\lambda T)^x}{x!}$ $\min_G \max_D L_{WGAN}(G, D) + \eta L_{AE}(G)$		

Table 1 (continued)

Study	Training ground-truth	Network details	Input	Output	Noise/de-noising model	Performance	Limitations
Klug et al. (2022)	Normal-dose scans	Posed-based convolutions Stacked convolutional auto-encoders	55 x 55 patches of noisy image	De-noised image	NA	PSNR: 47.00 SD: 29.9	- Small dataset - No categorization for CACS was evaluated - Low performance with calcification cases - Severe obesity cases were not evaluated
Jeon et al. (2023)	High-dose scans	Advanced Intelligent Clear-IQ Engine (commercial tool)	512x512 noisy image	De-noised image	NA	SSIM: 0.938 PSNR: 24.50 SD: 22.1	- Small dataset - Vendor specific algorithm - Extremely-low quality images were excluded - No disease cases were evaluated
Kobayashi et al. (2023)	Synthetic de-noising data using 4D noise reduction (4DNR)	Residual dense network (RDN)	48x48 patches of noisy image	De-noised image	NA	PSNR: 37.20 SD: 15.2	- 4D can not be applied to high heart rate cases - No noise power spectrum analysis
Orii et al. (2023)	Ultra-high resolution scans	Super-resolution convolutions	512x512 noisy image	De-noised image	NA	PSNR: 16.30 SD: 21.00	- Small dataset - Vendor specific algorithm - No severe CACS cases were evaluated
Generative adversarial network (GAN)							

Table 1 (continued)

Study	Training ground-truth	Network details	Input	Output	Noise/de-noising model	Performance	Limitations
Yang et al. (2019)	Normal-dose scans	Generator: U-Net Discriminator: U-Net	512x512 high-frequency and low-frequency noisy images	Generator: $x_h = \sigma_h(y_h)$ Noise pattern: $x_l = \sigma(y) - \sigma_h(y_h)$ Discriminator: $L_D(G) = E_{x,y,z}[\ Y_H - \hat{Y}_H\ _{L1}]$ Discriminator: $L_G(G) = E_{x,y,z}[\ Y - \hat{Y}\ _{L1}]$	$x_h = \sigma_h(y_h)$ $x_l = \sigma(y) - \sigma_h(y_h)$ $L_D(G) = E_{x,y,z}[\ Y_H - \hat{Y}_H\ _{L1}]$ $L_G(G) = E_{x,y,z}[\ Y - \hat{Y}\ _{L1}]$	SSIM: 0.899 PSNR: 31.27	- No external validation - No severely noisy cases
Kim et al. (2020)	Normal-dose scans	Generator: ResNet Discriminator: CNN	64x64 patches of noisy image	Generator: $L_G = L_{filter} + \alpha L_{L2} + \beta L_{structural}$ Noise pattern: $L_D = L_D + \alpha L_{L2} + \beta L_{structural}$ Discriminator: $L_G(G) = E_{x,y,z}[\ Y - \hat{Y}\ _{L1}]$	$L_G = L_{filter} + \alpha L_{L2} + \beta L_{structural}$ $L_D = L_D + \alpha L_{L2} + \beta L_{structural}$ $L_G(G) = E_{x,y,z}[\ Y - \hat{Y}\ _{L1}]$	SSIM: 0.922 PSNR: 39.92 RMSE: 0.928	- Small dataset - No external validation - No local dataset with original noise
Ma et al. (2020)	Normal-dose scans	Generator: ResNet Discriminator: CNN	55x55 patches of noisy image	Generator: $x = \sigma(y)$ Noise pattern: $\arg \min_f \ f(x) - y\ _2^2$ Discriminator: $L = \alpha L_{LSGAN} + \beta L_{SSIM} + \gamma L_{L1}$	$x = \sigma(y)$ $\arg \min_f \ f(x) - y\ _2^2$ $L = \alpha L_{LSGAN} + \beta L_{SSIM} + \gamma L_{L1}$	SSIM: 0.858 PSNR: 29.40 RMSE: 13.54	- Small dataset - No external validation - No local dataset with original noise

Table 1 (continued)

Study	Training ground-truth	Network details	Input	Output	Noise/de-noising model	Performance	Limitations
Gu et al. (2021)	Wavelet de-noised image	Cycle structure Generator: U-Net Discriminator: CNN	128x128 patches of High-frequency sub-band image	Generator: Noise pattern discriminator: denoised image	$L_{total}(G, F, D_X, D_Y) = -L_{GAN}(G, F, D_X, D_Y) + \lambda_{cycle} L_{cycle}(G, F) + \lambda_{identity} L_{identity}(G, F)$	SSIM: 0.624 PSNR: 29.89	- Ground-truth was generated - No extreme low-dose noise cases
Usui et al. (2024)	Full projection scan	Generator: U-Net Discriminator: CNN	512x512 noisy image	Generator: Noise pattern discriminator: crimi-nator: denoised image	$L_{cGAN}(G, D) = E_{x,y}[\log D(x, y)] + E_{x,z}[\log(1 - D(x, G(x, z)))]$	SSIM: 0.880 PSNR: 30.00	- Distortions in tumor structures - Noise were simulated from projections
U-Net							
Jing et al. (2022)	Normal-dose scans	Self-attention blocks	512x512 noisy image	Denoised image	$\hat{x} = f(x)$ $\min_{\phi} E_{x,y}[L(f_{\phi}(x), y)]$	SSIM: 0.915 PSNR: 29.07	- Small dataset - No external validation - No local dataset with original noise
Sandfort et al. (2024)	Synthetic de-noised data using 3D median filtering	3D structure	128x128x16 noisy image	Denoised image	NA	PSNR: 70.10 SD: 5.00%	- Training on synthetic de-noised data

Table 1 (continued)

Study	Training ground-truth	Network details	Input	Output	Noise/de-noising model	Performance	Limitations
Diffusion models							
Xia et al. (2022)	Normal-dose scans	Ordinary differential equation (ODE) solver U-Net structure	256x256	De-noised image	$q(y_t y_0) = N(y_t \sqrt{\alpha_t}y_0, (1-\alpha_t)I)$ $p_\phi(y_{t-1} y_t, x) = N(y_{t-1} \mu_\phi(y_t, x, t), \sigma_t^2 I)$	SSIM: 0.972 PSNR: 42.73	- Small dataset - No external validation - Simulated Gaussian noise with no reference
Gao et al. (2023)	Normal-dose scans	Multi-layer perceptron (MLP) Error-modulated module (EMM) U-Net structure	512x512x3 noisy images	De-noised image	$X_t = D(X_0, X_T, t) = \sqrt{\alpha_t}X_0 + \sqrt{(1-\alpha_t)}X_T$	SSIM: 0.958 PSNR: 40.71 RMSE: 18.6	- Simulated noise does not reflect real noise - High computational time and memory - Clinical testing was not sufficient
Liu et al. (2023)	Normal-dose scans	Conditional modelling Cascaded residual U-Net	512x512 noisy images	De-noised image	$p_\phi(X_{t-1} X_t) = N(X_{t-1}; \mu_\phi(X_t, t), \sigma_t^2 I)$ $p_\phi(X_{t-1} X_t, c) = N(X_{t-1}; \mu_\phi(X_t, t, c), \sigma_t^2 I)$	PSNR: 35.29	- Small dataset - No external validation - Size of parameters and lengthy processing time - Down-sampling may introduce degradations
Wang et al. (2024)	Normal-dose scans	Residual U-Net	128x128 patches of noisy image	De-noised image	$q(X_t X_0, X_1) = N(X_t; \frac{\sigma_t^2}{\sigma_t^2 + \sigma_t^2}X_0 + \frac{\sigma_t^2}{\sigma_t^2 + \sigma_t^2}X_1, \frac{\sigma_t^2 \sigma_t^2}{\sigma_t^2 + \sigma_t^2}X_0 I)$	SSIM: 0.950	- Small dataset - No external validation - No local dataset with original noise

Moreover, several methods are available to characterize the effect of the blooming artifact on the reconstructed image. For example, the HU values can be used to calculate the contrast-to-noise (CNR) ratio around a region of interest (ROI):

$$CNR = \frac{HU_{ROI} - HU_{background}}{\sigma_{HU_{background}}}, \quad (6)$$

where $\sigma_{HU_{background}}$ is the standard deviation of the HU values in the background.

In addition, the SNR can be calculated as:

$$SNR = \frac{HU_{ROI}}{\sigma_{HU_{background}}} \quad (7)$$

According to these equations, noise can be summarized as the $\sigma_{HU_{background}}$.

A widely used measurement for the length of the calcified region is the full width at third maximum (FW3M) Xu et al. (2019), which allows for the calculating of the adjacent pixels affected by blooming (A_n):

$$A_n = l_{bright} - l_{actual}, \quad (8)$$

where l_{bright} is the length of the bright pixels around the calcified region and l_{actual} is the actual calcification length. Another way is to calculate the percentage error caused by blooming when reconstructing a CCTA image in the axial view Park et al. (2024). Here, the distortion angle (ϕ_d) in degrees can be calculated:

$$\phi_d = \frac{180}{\pi} \times \arctan \sqrt{(\tan \phi_1)^2 + (\tan \phi_2)^2}, \quad (9)$$

where ϕ_1 and ϕ_2 are the distortion angles in the anterior and left views.

Next, the total error ($Error_t$, %) in measurement can then be calculated as:

$$Error_t, \% = \left(1 - \frac{1}{\cos \phi_d}\right) \times 100 \quad (10)$$

Identifying the calcified region and measuring the calcified volume (CV), diameter of stenosis (DS), and area of stenosis (AS) before and after applying any de-blooming approach is also of interest in several studies Li et al. (2018). This allows for calculating the following parameters:

$$RCV = \frac{(CV_{before} - CV_{after})}{CV_{before}}, \quad (11)$$

$$RDS = \frac{(DS_{before} - DS_{after})}{DS_{before}}, \quad (12)$$

$$RAS = \frac{(AS_{before} - AS_{after})}{AS_{before}}, \quad (13)$$

where RCV, RDS and RAS are the relative reductions in calcification volume, coronary diameter stenosis, coronary area stenosis reduction over calcified plaque, respectively.

2.2.2 Conventional approaches to de-noising

Conventional approaches to restore image quality due to the blooming effect include utilizing stronger X-ray beams with higher energy levels (such as kVP or keV) are suggested Stocker et al. (2020); however, this may be harmful to the tissues being imaged, as discussed in the previous section. In addition, this approach usually fails to reduce the number of adjacent pixels expected to be affected by this artifact Park et al. (2024). Alternative solutions including increasing the window width (grey-level mapping), reducing the field-of-view (FOV), applying a sharp kernel with the use of the IR algorithm, and reducing slice thickness Triche et al. (2019). In general, these methods would require manual selection, tuning, or applying assumptions on the expected size and location. On the other hand, several studies utilized conventional post-processing approaches applied directly to images to correct for the presence of blooming artifacts. For example, a method relying on photon counting was used to improve vasa vasorum detection in CT images of porcine models Marsh et al. (2020) and ex-vivo human cadavers coronary specimens VanMeter et al. (2022). However, this technique is not ideal when exposed to high intensity X-ray Taguchi and Iwanczyk (2013), e.g., during vascular imaging with a contrast agent. A variety of filtering methods have been also used including material decomposition (Li et al. 2020c), decomposition-based iterative reconstruction (Do et al. 2011), iterative deconvolution (Hofmann et al. 2014), iterative correction (Li et al. 2018; Weir-McCall et al. 2020), and bilateral filtering (Steckmann and Kachelrieß 2010). Despite their efficiency in removing the blooming artifacts, they remain sensitive to the initialization criteria, i.e., material properties or choice of initial parameters. Moreover, due to their iterative nature, these methods tend to suffer from convergence issues, while being computationally intensive.

2.2.3 Deep learning approaches to de-noising

Most recently, deep learning algorithms have also been investigated for post-processing and de-blooming CT and CCTA images (Table 2). To address this, CNN models were used during the reconstruction of CCTA with a significant reduction in the blurry appearance of calcified plaques (Higaki et al. 2020; Yi et al. 2021; Otgonbaatar et al. 2021). In Yi et al. (2021), the authors proposed a reconstruction algorithm based on CNN for CCTA noise removal taking into consideration invasive coronary angiography (ICA) as the ground truth. Their approach reported a SNR of 27.24 dB, however, heart motion and other vessel caliber changes were not considered. U-Net models have been widely implemented in several studies Zhang and Xing (2018), Agrawal et al. (2023), Chang et al. (2023), owing to their ability to fuse features identified at different layers, while excluding blooming-related artifacts. In Chang et al. (2023), the model utilized prior information about the noise components and outperformed FBP by $95 \pm 1\%$ and IR by $60 \pm 8\%$. However, expert assessment is needed to elaborate on its clinical validity, i.e., calcium scoring. An advanced model based on GANs

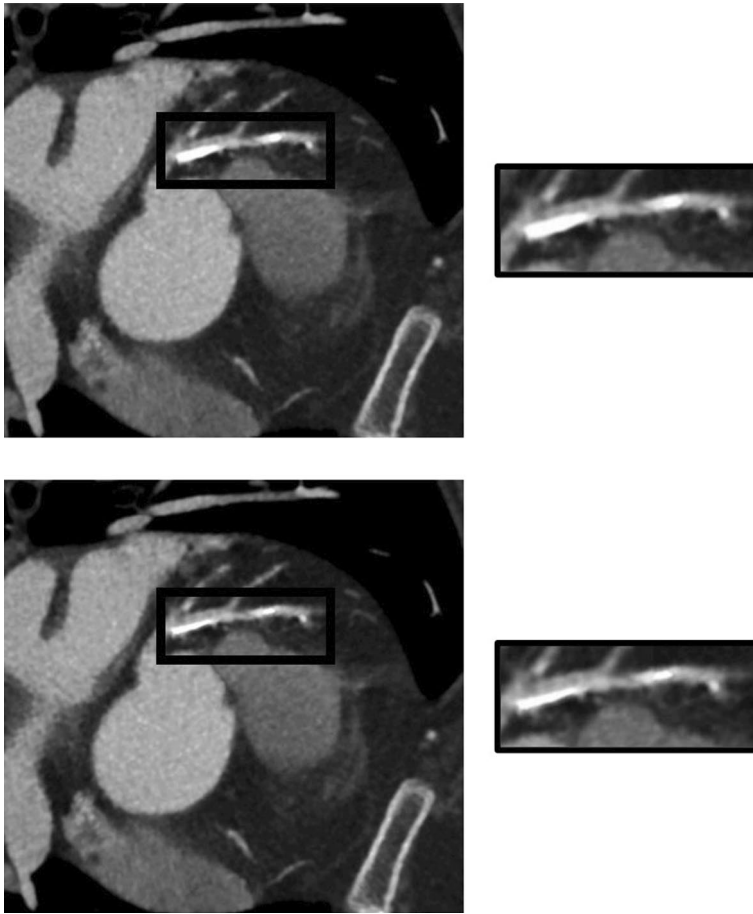


Fig. 2 Example of blooming artifact caused by calcification taken from a CCTA image. The example is for a CCTA slice with a deformed calcified plaque caused by the partial volume averaging (Pack et al. 2022) (upper) and the de-bloomed CCTA slice through iterative correction (Wuest et al. 2013) (bottom); showing closer size to the true one. Retrieved from Weir-McCall et al. (2020) with additional editing

was proposed in CCTA calcium de-blooming (Sun and Ng 2022a, 2022), resulting in high-performance levels and an overall increase in the diagnostic value of coronary calcification. Here, the model reached up to 81.3% and 80.0% in sensitivity and specificity, respectively, in identifying calcified plaques with a de-noising performance as low as $10.99 \pm 13.94\%$ standard deviation from the original image. This performance was not equally distributed at different parts within the arteries, i.e., LAD, especially in the presence of severe calcification.

2.3 Beam hardening

2.3.1 Background and nature

Beam hardening is a phenomenon that occurs in CT imaging as the X-ray energy increases while the attenuation coefficients for human body tissues decrease (Albrecht et al. 2018).

Here, the polychromatic X-ray beam penetrates through the object of interest (OI) with all its tissues causing more attenuation for lower energy photons, leading to an increase in the mean energy of the transmitted beam (Bucher et al. 2016). This increase leads to “hardening” the beam and closely resembles a high-pass filter that allows high energy photons only to contribute to the X-ray beam (Tesche et al. 2017). At lower kV levels, this phenomenon leads to a difference between the actual and assumed attenuation levels of the HU voxels, which affects the reconstruction of CT images He et al. (2015). Beam hardening results in non-linear artifacts in the generated CT image (Fig. 3), i.e., streaking effect and cupping artifacts Ahmed and Song (2018). Streaking effects appear due to hardening the beam at different rates while changing the rotation and position of the X-ray tube Imai et al. (2009). It resembles dark streak bands that show up between two dense objects, i.e., bones, iodinated contrast, metals, and other high-attenuation materials that create bright lines that overlap with darker regions in the image Gjestebj et al. (2019). On the other hand, the cupping artifact appears for longer paths within the image, i.e., when scanning phantoms (Hunter and McDavid 2012; Nagarajappa et al. 2015), and is caused by the scattering flux when the linear attenuation coefficient is underestimated (Rinkel et al. 2007). In current practice, it rarely appears in CT imaging due to advancements in modern scanners. However, it still creates issues when imaging other OI Zeng (2022a). Considering projection, photons carrying less energy are highly attenuated over a long scanning path, causing the middle of the image to decrease in voxel value, e.g., a uniform cylindrical object would appear with reduced grey level values more towards its center Ahmed and Song (2018).

In CCTA, beam hardening appearance closely resembles images taken for patients with myocardial ischemia (Carrascosa et al. 2015), i.e., having a streak- or band-like noise shape within the image. These artifacts may cause difficulties when assessing the inferior left ventricle for hypoperfusion (Albrecht et al. 2018). For example in cardiac imaging, some areas are more exposed to such artifacts, i.e., areas between dense anatomical structures, namely the contrast-enhanced left ventricle (LV), thoracic aorta, and other areas near the spine (Kitagawa et al. 2010; Bucher et al. 2016).

2.3.2 Mathematical model

In CT, the X-ray beam is generated as a polychromatic wave with an exponential attenuation that is governed by the Beer-Lambert’s law Seeram (2010):

$$I = I_0 e^{-\mu L}, \quad (14)$$

where I is the intensity of the recorded X-ray beam, I_0 is the intensity of the transmitted X-ray beam, μ is the linear coefficient of attenuation for the material being screened, and L is the length of the path that the X-ray beam travels through. Due to the nature of the screened object, which usually contains tissues with varying properties, the linear attenuation should be considered in terms of material type (E) and length of the traveling path (L) as:

$$I = I_0 e^{-\int \mu(E)L(E)dE} \quad (15)$$

In most cases, accurate modeling of the polychromatic X-ray beam is not feasible due to material type, the voltage of acceleration, and filtration within the CT machine (Ketcham and Hanna 2014; Hanna and Ketcham 2017). Therefore, the X-ray beam is often assumed to have a monochromatic energy nature with no consideration for the full X-ray spectrum (Ahmed and Song 2018). This assumption during the reconstruction of CT images causes the beam hardening artifact.

In the context of beam hardening, the cupping artifact (C) can be modelled as:

$$C = \left[\frac{(a + \frac{b}{2}) - m}{m} \right] \times 100, \quad (16)$$

where a and b are the maximum values for the gray levels within the attenuated region and m is the minimum gray level value at the center of the cupping region (Arunmuthu et al. 2013; Ketcham and Hanna 2014).

2.3.3 Conventional approaches to de-noising

To correct the beam hardening artifacts in CT imaging, studies have often utilized hardware- and software-based methods. The main objective is to reconstruct monochromatic linear attenuation coefficients for the screen object given the transmitted polychromatic X-ray beam. Hardware-based methods depend on handling the transmitted X-ray beam characteristics, i.e., allowing high-energy rays to pass through Ahmed and Song (2018). For example, several studies have used filters to reduce the transmitted low-energy rays to the object (Meganck et al. 2009; Ay et al. 2013). The filter has characteristics such as being made from copper, aluminum, or iron, as well as having different thicknesses and SNR ratios (Chen et al. 2016; Ahmed and Song 2018). Overall, hardware-based methods cause a severe drop in the transmitted beam's intensity without proper removal of the artifacts.

In software-based methods, several studies have commonly implemented the Monte-Carlo simulation approach (Thomsen et al. 2015; Jin et al. 2015; Nazemi et al. 2021), which provides a direct experimental measure for the X-ray spectrum before screening the OI. Another method relies on the linearization of the attenuation measure relative to the thickness of the object (Vavrik and Jakubek 2009; Lifton 2017; Martinez et al. 2022), however, it requires samples with a limited number of materials with simple geometry.

In addition, the most commonly used method is empirical beam hardening correction (EBHC), which is often used to remove bone-induced streaks (Kyriakou et al. 2010). In this technique, the sinogram gets corrected to segment soft tissues and bones, obtaining only water pre-corrected images (Reiter et al. 2019; Andreyev et al. 2022). Lastly, IR is commonly used to account for the beam hardening artifacts by analyzing the spectral information and geometry of the system and OI to reduce the attenuation (Brabant et al. 2012). For example, Jin et al. (2015) proposed a model-based IR algorithm (MBIR-BHC) to model the beam hardening artifact with only knowledge of the screened materials. In general, IR algorithms require longer computational times, which is usually aligned with higher costs in clinical applications (Ahmed and Song 2018).

2.3.4 Deep learning approaches to de-noising

In recent studies (Table 3), several authors utilized deep learning models to correct for the beam hardening artifact, demonstrating better performance and applicability compared to conventional methods Njiti et al. (2024). For metal-induced beam hardening artifacts, CNN were proposed to reduce the noise and maintain the anatomical structure of the OI (Zhang and Yu 2018; Ziabari et al. 2020; Ghafarzadeh et al. 2024). For example, *DestreakNet* Gjestebj et al. (2018) was proposed by Gjestebj et al., to reduce streak artifacts in hip prostheses cases screened using CT, which prevented information loss compared to conventional methods. Despite achieving high performance with SSIM of 0.86 and PSNR of 23.86 dB, this approach was tested on phantom images with no clinical validation. Moreover, U-Nets were also investigated while incorporating knowledge about the structural properties of the object (Park et al. 2018; Bayaraa et al. 2020; Kalare et al. 2022). Most recently, Jiang et al. (2024) developed a combined model of U-Nets and ResNets to extract multi-scale features while correcting projection- and image-domains information. Their approach reached a PSNR of 36.13 dB and SSIM of 0.953, however, they proposed a computationally expensive model with no clinical testing. GANs were also utilized to handle this type of noise across multiple studies Ghani and Karl (2019), Khaleghi et al. (2021), Wang et al. (2021). In Wang et al. (2019), the authors proposed a GAN model on physical simulations of artifacts in CT images to correct for artifacts caused by metal cochlear implants. For bone-induced beam hardening artifacts, Ji et al. (2023) replaced the conventional EBHC empirical structure with a neural network (U-Net), enforcing a novel physics-driven de-noising approach in brain images. Applications in dental imaging were also investigated in several studies Bayaraa et al. (2020), Martínez et al. (2021) mostly relying on a U-Net structure to segment the artifact effect from the image. It is worth noting that beam-hardening artifact removal using deep learning received minimal attention in the area of cardiac imaging and CCTA.

Fusion of FBP and deep learning was proposed in Olasz et al. (2021) to remove beam hardening artifacts, while maintaining structural consistency, essential in coronary assessment. A comprehensive two-stage framework was proposed in Bayaraa et al. (2020) for low-dose dental CBCT imaging. Sinogram adjustment is performed to improve data consistency, while reducing beam hardening artifacts, followed by CNN processing for noise suppression, reporting improvements in SSIM. Super-resolution deep learning reconstruction [3] was shown to reduce calcification-related blooming, while improving vessel representation, indirectly reducing polyenergetic beam-hardening. In summary, the majority of existing deep-learning based beam hardening denoising methods are trained on simulated or specialized acquisition data to learn the representations of artifact-free instances. In regards to the reported quantitative results, these originate on phantom-based studies and non-cardiac scans, thus lacking evidence on true performance in the case of coronary arteries. While the majority of vendor DLR provide improvements in overall image quality, beam hardening suppression has not been systematically evaluated in the coronary setting. Deep learning denoising approaches result in improved vessel delineation, however, they may result to over-smoothing, potentially altering calcium density Nagayama et al. (2023). CNNs have been successfully used in controlled settings to reduce beam hardening streaks, however, additional research is needed to ensure reliable CCTA correction.

Table 2 Summary of the state-of-the-art studies on the use of deep learning for blooming artifact correction

Study	Year	Modality	Organ	Image dimension	Dataset	Blooming origin	
Convolutional neural network (CNN) Otgonbaatar et al. (2021)	2021	CTA	Brain	2D	43 patients (Local dataset)	Metal objects	
	2021	CCTA	Heart	2D	42 patients (Local dataset)	Calcified plaques	
Generative adversarial network (GAN) Sun and Ng (2022a)	2022	CCTA	Heart	2D	50 patients (Local dataset)	Calcified plaques	
	2022	CCTA	Heart	2D	50 patients (Local dataset)	Calcified plaques	
U-Net Sun and Ng (2022a)	2022	CCTA	Heart	2D	10 patients (Local dataset)	Calcified plaques	
Agrawal et al. (2023)	2023	CT	Brain	2D	20 scans (Local dataset)	Metal objects	
Study	Comparison reference	Network details	Input	Output	De-blooming model	Performance	Limitations
Convolutional neural network (CNN) Otgonbaatar et al. (2021)	High-dose MBIR/FBP	Advanced Intelligent Clear-IQ Engine (commercial tool)	Images with blooming	De-bloomed images	NA	SNR: 30.27 CNR: 34.28 Quality score: 3.5	- Small dataset - Patient with no pathology - No consideration for radiation exposure - No comparison with other methods

Table 2 (continued)

Study	Comparison reference	Network details	Input	Output	De-blooming model	Performance	Limitations
Gen-erative ad-verse network (GAN)	Yi et al. (2021)	Advanced Intelligent Clear-IQ Engine (commercial tool)	Images with blooming	De-bloomed images	NA	SNR: 27.24 CNR: 33.20	- Small dataset - No vessel caliber changes - Subtraction without con-sideration for heart motion
	Sun and Ng (2022a)	Generator: CNN Discriminator: CNN	512x512 with blooming	2048x2048 super-resolution de-bloomed images	$D(x) = \sigma(C(x))$ $L_D = -E_{x_r}[\log(D(x_r, x_f)) - E_{x_f}[\log(1 - D(x_f, x_r))]]$ $L_G = -E_{x_r}[\log(1 - D(x_r, x_f)) - E_{x_f}[\log(D(x_f, x_r))]]$	Specificity: 55.00% ROC: 0.80	- Different CT scanner types - models trained on non-CCTA images
	Sun and Ng (2022b)	Generator: CNN Discriminator: U-Net	512x512 with blooming	2048x2048 super-resolution de-bloomed images	$D(x) = \sigma(C(x))$ $L_D = -E_{x_r}[\log(D(x_r, x_f)) - E_{x_f}[\log(1 - D(x_f, x_r))]]$ $L_G = -E_{x_r}[\log(1 - D(x_r, x_f)) - E_{x_f}[\log(D(x_f, x_r))]]$	Sensitivity: 81.3% Specificity: 80% ROC: 0.85	- Different CT scanner types - Low performance in LAD and LCx - Low performance in high calcification - No analysis between calcified and non-calcified regions

Table 2 (continued)

Study	Comparison reference	Network details	Input	Output	De-blooming model	Performance	Limitations
U-Net							
Sun and Ng (2022a)	70 keV image	Exponential linear unit (eLU) activation	128x128 noisy patches	De-bloomed patches	$L(\phi) = \frac{1}{M} \sum_{i=1}^M y^i - f(x^i, x_{prior}^i, \phi) ^2$	Reduction in blooming	- Small dataset - Clinician assessment was not included - Single image acquisition setting
Agrawal et al. (2023)	Simulated images	Basic convolutions	948x740 noisy images	De-bloomed images	$l = \frac{1}{N \times w \times h} \sum_{i=0}^{N \times w \times h} y_i \log(\sigma(x_i)) + (1 - y_i) \log(1 - \sigma(x_i))$	Reduction in blooming DSC: 94.8% in segmenting metals	- Small dataset - Only simulated data for training/testing

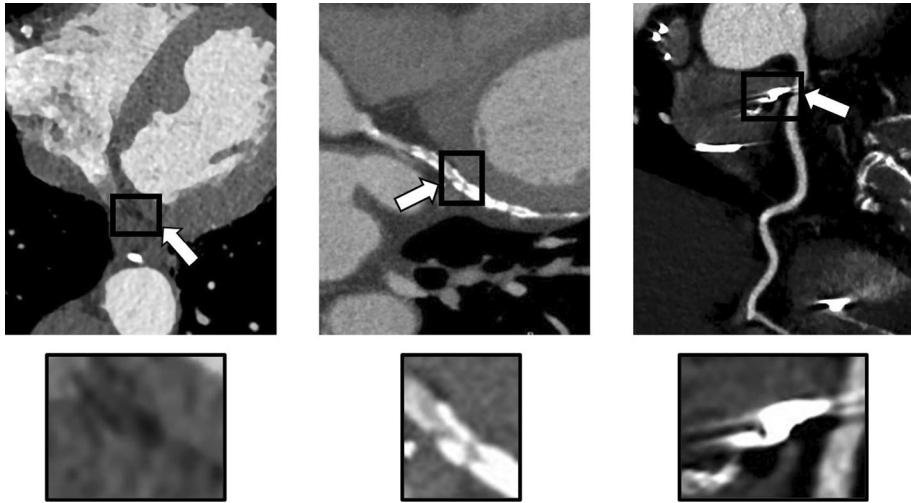


Fig. 3 Beam hardening artifact on images taken from coronary computed tomography angiography (CCTA). The example (left column) is for a CCTA slice with a streak-like artifact around the calcified region in the right coronary artery (RCA) causing hypodensity in the image. Another example (middle column) shows another streak artifact with low-density areas on the areas near the calcified plaque. The third example (right column) shows a beam hardening artifact in the right coronary artery caused by a pacemaker lead in the right atrium. Retrieved from Albrecht et al. (2018), Andrew and John (2015), and Sirajuddin et al. (2021), respectively, with additional editing

2.4 Motion artifacts

2.4.1 Background and nature

In CT imaging, patient-specific artifacts occur due to voluntary and involuntary movements, i.e., motion artifacts, in the course of image acquisition Kyme and Fulton (2021). During the imaging procedure, patients are requested to remain extremely still, as any small movement can cause disturbances in the generated images and thus, affect the diagnostic value and associated examination results Watkins et al. (2010). Voluntary motion typically arises from discomfort, anxiety, speaking, or chewing, which cause muscular movements of the head or body. Patients may perform more complex movements like twisting or bending that severely affect the screening quality Gu et al. (2010). On the other hand, involuntary movements, such as cardiac, respiratory, bowel, fetal, peristalsis, or body reflex motion, generate artifacts that appear as blurring or long-range streaks Boas and Fleischmann (2012). For example, respiratory-induced inspiration and expiration create artifacts in chest CT, and thus, breath-hold imaging protocols are advised to prevent the movement of the diaphragm. However, these scans are limited to less than 1 min, i.e., short scanning applications Kyme and Fulton (2021), compared to long-scan CT, where avoiding motion is difficult.

This kind of artifacts varies in strength due to many factors including motion type, scanning duration, gantry speed, and motion direction Zijlstra et al. (2017). For example, the type of motion creates complex distortions to the generated images that appear different compared to regular rigid, non-rigid, periodic, random, rotational, or drifting movements Geramifar et al. (2013). Various operating factors, e.g., gantry rotation, object positioning,

spatial/temporal resolution of the modality, as well as patient-specific parameters also affect the impact of motion on the generated image Xu et al. (2012). However, in modern CT modalities, the speed of the gantry (0.3 s/revolution) is fast enough to overcome significant motion in a single frame Boas and Fleischmann (2012).

Motion artifacts (Fig. 4) manifest themselves with varying appearance, e.g., arc-shaped blurring, intensity undershoots, harsh smearing, and ghost-like repetitions (Lossau et al. 2019). During motion, edges with higher contrast in the view between the object and the X-ray tube are mostly affected. This causes temporal discrepancies between different time points as they may not align properly within the reconstructed image. When bones or metallic objects are in the field-of-view of the scanner during the imaging session, motion appears as bright or dark lines near these high-contrast objects (Nagarajappa et al. 2015). This type of artifact causes the imaged object to have repeated shapes around the edges, i.e., ghosts, due to a shift in position during the scan (Sun et al. 2021), thus distorting the spatial positioning of the anatomical object within the generated image.

2.4.2 Mathematical model

Motion artifacts are caused by multiple sources and can be expressed as the effect of object displacement. The displacement motion within a single gantry rotation (ΔS) is represented with respect to time (t) as follows:

$$\Delta S = \int_{t-\delta/2}^{t+\delta/2} v(t) dt, \quad (17)$$

where δ is the duration of a single gantry rotation and $v(t)$ is the time-dependent target velocity.

In a normal scanning session, motion caused by involuntary actions, i.e., breathing, has a constant velocity at each phase (Nakamura et al. 2009). Here, the target motion (S) can be expressed as:

$$S(t) = A \cos(\omega t), \omega = 2\pi/T, \quad (18)$$

where A is the ideal periodic motion amplitude and T is the period.

In a single gantry rotation period, i.e., $t \pm \delta/2$, the sinusoidal displacement motion (ΔS) is:

$$\Delta S = |A \cos \omega(t + \delta/2) - A \cos \omega(t - \delta/2)| \quad (19)$$

or equivalently,

$$\Delta S = 2A \sin(\omega \delta/2) \sin(\omega t) \quad (20)$$

This formula explains the amount of blurring in the reconstructed slice, i.e., the displacement movement (ΔS), during one CT gantry rotation. To address the extremum phases during motion, e.g., strong exhalation or severe motion, the displacement motion can be considered linearly dependent on the target's velocity (Watkins et al. 2010):

$$\Delta S = \frac{2}{\omega} \sin(\omega\delta/2) |v| \quad (21)$$

Once the velocity reaches zero, i.e., peak exhalation, the displacement can be calculated as the difference between the initial position and the extrema:

$$\Delta S = (A - |A \cos \omega(t \pm \delta/2)|), v = 0 \quad (22)$$

When evaluating for $v = 0$ or $\omega t = n\pi$ for $n = 0, 1, 2, \dots$ etc, the total displacement becomes:

$$\Delta S = A(1 - \cos(\omega\delta/2)) = 2A \sin^2(\omega\delta/4) \quad (23)$$

Finally, the effect of the motion-induced noise is bounded by Eqs. 20 and 23 for initial displacement and extreme motion, respectively.

2.4.3 Conventional approaches to de-noising

To mitigate motion artifacts that can compromise diagnostic accuracy, clinicians employ various techniques centered on optimizing CT imaging protocols. Modern CT scanners are designed to reduce scan duration, minimizing patient movement and subsequently reducing the likelihood of motion artifacts. Ko et al. (2021). For example, rapid scanning techniques are particularly beneficial in cardiac imaging, allowing for the capture of the heart and coronary arteries during diastole within a single heartbeat, thereby enhancing image clarity and diagnostic confidence. ECG gating is another clinically significant approach that synchronizes image acquisition with specific phases of the cardiac cycle (Athanasios et al. 2017), allowing the reconstruction of 3D volumes with reduced motion artifacts, which is critical for accurate coronary artery evaluation and intervention planning. However, both approaches often expose the patient to higher doses or longer scanning time which increases associated risks of X-ray beams. Advanced cardiac scanners with an increased number of detector rows can further enhance imaging by covering larger volumes in a single rotation, effectively reducing step-off motion artifacts and improving diagnostic reliability. These innovations are pivotal in clinical practice, where precise imaging of coronary anatomy directly impacts patient management decisions. Yu and Wang (2007), Boas and Fleischmann (2012). However, incorporating machines with a larger number of detectors increases scan cost, machine maintenance requirements, and complexity of the hardware equipment.

Other approaches rely on compensating for patient motion through prior estimation. Direct and indirect motion estimation was utilized to exclude the motion pattern during CT scans. For example, direct motion tracking systems, i.e., cameras and markers, were implemented to compensate for motion during the reconstruction phase (Choi et al. 2014; Kim et al. 2015). Despite their success in reducing motion effect, they still poses limitations concerning costs, calibration, and continuous maintenance and configuration. In addition, they depend on the ability of the system to interpret patient motion and cameras recording; which are often unpredictable. Indirect methods rely on determining motion parameters based on patient information, i.e., heart rate, without the need for additional equipment (Tang et al.

Table 3 Summary of the state-of-the-art studies on the use of deep learning for reducing beam-hardening artifacts

Study	Year	Modality	Organ	Image dimension	Dataset	Noise origin
Convolutional neural network (CNN) Zhang and Yu (2018)	2018	CT	Lungs	2D	74 metal-free - 31 metal subjects (CT Grand Challenge McCollough (2016)) 1 patient (clinical data)	Metal
	2018	CT	Hips	3D	50 phantom image pairs (Visible Human Project Ackerman (1998))	Metal
Generative adversarial network (GAN) Khaleghi et al. (2021)	2021	CT	Teeth	2D	2,600 metal-free - 2,600 metal (Simulated images) 7,000 metal-free - 7,000 metal subjects (Local dataset - 30 patients)	Metal
	2021	CT	Ear	3D	1,000 temporal bone images (Cochlea dataset - 597 patients)	Metal
U-Net Bayarar et al. (2020)	2020	CT	Teeth	3D	658 slices (Local dataset - phantom)	Bone
Martínez et al. (2021)	2021	CT	Teeth	3D	375 projections (Local dataset - 30 rodents)	Bone
Ji et al. (2023)	2023	CT	Brain	2D	4,906 slices (Local dataset - 30 patients)	Bone

Table 3 (continued)

Study	Comparison reference	Network details	Input	Output	Noise/de-noise model	Performance	Limitations
Convolutional neural network (CNN)							
Zhang and Yu (2018)	Metal-free scans	2D CNN	10,000 patches (64 × 64)	Noise-free image	$x^{CNN} = C_L(x^{input})$ $\bar{x}^{CNN,w} = \frac{\sum_i D_i x_i^{CNN}}{\sum_i D_i}$ $x_i^{prior} = \frac{D_i}{N} \bar{x}^{CNN,w} + (1 - \frac{D_i}{N}) x_i^{CNN}$	SSIM: 0.97 RMSE: 30.1	- Manual selection of metal-free and metal images - Inaccurate metal segmentation affects the performance - Phantom images with no validation on clinical data - Simple neural network structure
Gjesteby et al. (2018)	Metal-free scans	2D CNN (DestreakNet)	150,000 patches (56 × 56)	Noise-free image	$I_{detail} = I - I_{base}$ $\hat{Y} = h(f(I) + g(I_{detail}))$ $L_{MSE} = \frac{1}{N} \sum_{i=1}^N \ \hat{Y}_i - Y_i\ ^2$	SSIM: 0.86 PSNR: 23.86	

Study	Comparison reference	Network details	Input	Output	Noise/de-noise model	Performance	Limitations
Generative adversarial network (GAN) Khaleghi et al. (2021)	Metal-free scans	Generator: U-Net Discriminator: CNN	256×256 images	Noise-free image	NA	RMSE: 17.68 PSNR: 23.50 CNR: 5.04	- Model fine-tuning performed on simulated data - Small clinical dataset - No deep details on the GAN network

Study	Comparison reference	Network details	Input	Output	Noise/de-noise model	Performance	Limitations
Wang et al. (2021)	Metal-free scans	Generator: U-Net Discriminator: CNN	512×512 images	Noise-free image	$L_D = E_x \log(D_{wd}(x)) + E_y \log(1 - D_{wd}(G_{wg}(y)))$ $L_G = \alpha L_{retinex} + L_{mse} + L_{adv}$	SSIM: 0.64 RMSE: 0.12 PSNR: 18.31	- Model fine-tuning performed on simulated data - Difficulty in implementing the non-linearity in simulations - No evaluation of the cochlear implant characteristics

Study	Comparison reference	Network details	Input	Output	Noise/de-noise model	Performance	Limitations
U-Net Bayaraa et al. (2020)	Artifact-free image	Basic structure	800 × 800 × 400 images	Noise-free image	$f_{dl} = \operatorname{argmin} \sum_{k=1}^N \sum_{j=1}^M f_{dl}(\mu_{\#}^{(k,j)} - \mu_{*}^{(k,j)}) _2^2$	SSIM: 0.97	- Simulated beam hardening effect - No clinical dataset - Basic U-Net architecture
Martínez et al. (2021)	Artifact-free image	NA	512 × 512 × 375 images	Noise-free image	NA	NA	- Small dataset with no clinical data - No quantitative evaluation was performed - Basic computational model

Study	Comparison reference	Network details	Input	Output	Noise/de-noise model	Performance	Limitations
Ji et al. (2023)	Artifact-free image	Segmentation network Combination network	NA	Segmented image Noise-free image	$L_{mse} = \frac{ I_{ref} - I_{net} _2^2}{N}$	SSIM: 0.74 MAE = 8.21	- Simulated beam-hardening effect - Detector covers the whole object (data truncation)

2012; Zhang et al. 2013). However, they are patient-dependent and can be highly affected by abnormalities, e.g., arrhythmias.

Image-based techniques rely on iteratively updating motion displacement through image registration during reconstruction (Sun et al. 2016; Chee et al. 2019). These techniques create a prior motion model, particularly in the case of 4D cone-beam CT, using analytic, iterative, and algebraic methods Dhou et al. (2015), Guo et al. (2019), Dhou et al. (2022). However, most previous studies assume a regular motion pattern with no intercycle variations. Other studies proposed simultaneous motion estimation and image reconstruction (SMEIR), introducing target volume regularization on phase-binned cone-beam CT (Wang and Gu 2013; Huang et al. 2018b; Zhou et al. 2021). The deformation field gets estimated for all projections during forward and back-projection steps, which is computationally demanding. Additionally, the simultaneous algebraic reconstruction technique (SART) was proposed to handle motion artifacts owing to its ability to trace ray sums across the image for each gantry projection angle (Pengpan et al. 2012; Chee et al. 2019), however, it is considered slower in speed compared with other techniques. Statistical estimation for motion has also been developed using the covariance matrix adaptation evolution strategy (CMA-ES) for optimization and adjusting for image sharpness and motion smoothness trajectory during the reconstruction (Klima et al. 2016; Sisniega et al. 2017). A major disadvantage of this technique lies in that it requires continuous evaluation of an objective function, which often leads to increases in processing times and computational demands.

In CCTA, numerous image processing techniques have been applied to handle motion artifacts impacting the reconstruction of coronary arteries (Ghekiere et al. 2017). For example, iterative motion adjustments based on motion vector field (MVF) estimation was implemented to decrease motion artifact measures (MAMs) (Rohkohl et al. 2013). Tracking coronary arteries in three dimensions during multiple cardiac phases was used to correct for

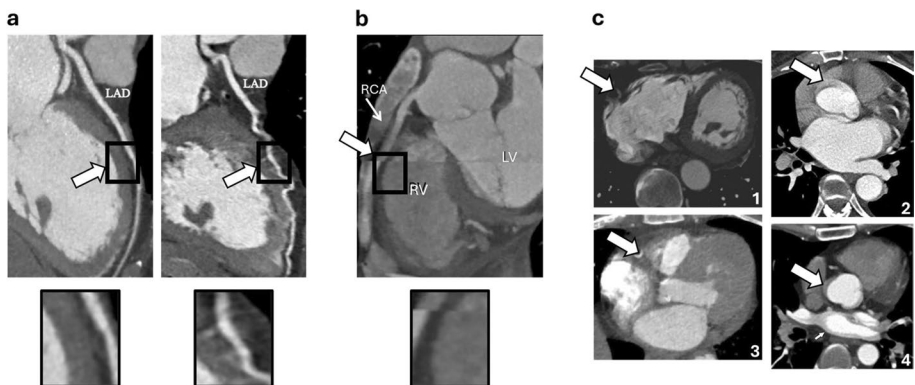


Fig. 4 Examples of motion artifacts taken from coronary computed tomography angiography images. **a** The example shows a 61 years old female patient with no motion artifacts (left) and a 59 years old female patient with substantial motion artifacts (right) affecting the left anterior descending artery (LAD) both with a heart rate of 68 bpm, **b** An example of motion artifacts affecting the proximal and mid segments of the right coronary artery (RCA) in a patient with a heart rate of 71 bpm, **c** multiple examples of cardiac motion artifacts with (1) blurring effect at the right coronary artery, (2) ghosting of the ascending aorta, (3) winging appearance of the right coronary artery, and (4) stair-step artifact at the aortic root. The examples are retrieved from Vecsey-Nagy et al. (2022), Bartykowszki et al. (2015), and Kalisz et al. (2016), respectively, with additional editing

intra-cycle motion artifacts, e.g., SnapShot Freeze (Liang et al. 2018; Wen et al. 2018; Liang et al. 2019; Yamaguchi et al. 2024). However, in presences of diseases such as arrhythmias, these techniques could be challenging. Multiple studies investigated another approach based on 3D-3D nonrigid image registration and showed a strong potential in handling motion displacement (Bhagalia et al. 2012; Jeong et al. 2023), however, registration approaches could be slightly erroneous and dependent on the initialization and algorithm (Ren et al. 2022).

2.4.4 Deep learning approaches to de-noising

Several deep learning models have been developed to tackle artifacts generated due to voluntary or involuntary motion in CT images (Table 4). Conventional CNNs were investigated in multiple studies to adjust for streak artifacts appearing due to breathing motion (Madesta et al. 2020; Sisniega et al. 2020; Zhang et al. 2023b; Amirian et al. 2024) in lung 4D CBCT imaging. For example, a prior-guided CNN was developed using prior high-quality cone-beam CT images, which serve as input image channels for the CNN model Zhi et al. (2019), Zhi and Mou (2021). However, the performance is dependent on the quality of the prior images, which could be affected by low resolution or artifacts themselves. CycN-Net was introduced to adjust phase-resolved noise occurring due to respiratory motion in 4D cone-beam CT using temporal correlation between scans (Zhi et al. 2021). The performance of the approach may be affected in situations, where scans are not sequentially valid, i.e., not sufficiently correlated due to rapid patient movements or irregular breathing patterns. Thummerer et al. (2022) developed a CNN-based model for estimating synthetic CT scans from time-resolved 4D cone-beam CT. Despite the promising performance, the authors stated that their models suffered from absence of ground-truth and clinical validation of its applicability during adaptive proton therapy workflows.

Studies have also implemented U-Net architectures, i.e., up-sampling and down-sampling, to handle artifacts caused by motion displacement (Lee et al. 2019; Madesta et al. 2019; Huang et al. 2020b; Jiang et al. 2021; Gunawan et al. 2022). The majority of these studies focus on respiratory motion in lung 4D CBCT imaging. For example, a simultaneous motion estimation and image reconstruction (SMEIR) network was developed based on a 4D U-Net structure to handle biomechanical and physical motion in intra-lung deformation vector fields (DVFs) prediction (Zhang et al. 2019; Huang et al. 2020b). However, this approach relies on images of the entire chest, i.e., including the diaphragm, to estimate DVFs, which could be limited during regular scans. Lastly, GANs were also tested in motion artifact correction and showed higher performance levels across multiple studies Madesta et al. (2019), Zhang et al. (2022), Dong et al. (2022). Here, authors utilize 4D CBCT images to correct for streak artifacts caused by respiratory motion using modifications of generative models, e.g., pix2pix. While being effective in noise reduction, they potentially require patient-specific information and image-related radiomics features. Moreover, Dong et al. (2022) utilized a generative approach on 2D images during training with no consideration for the temporal correlation between slices. A combination of GAN and FBP methods was proposed to handle motion artifacts using progressive growing, which creates high-resolution photo-realistic images Wang et al. (2020). This was tested on brain CT imaging with

voluntary motion and achieved high performance. However, it still lacks in considering real motion noise rather than relying on simulation models.

In CCTA, broader deep-learning applications were explored to handle motion artifacts in coronary artery images. In particular, CNN models were implemented in several studies to train the model for heart displacement caused by breathing and heart motion (Elss et al. 2018; Jung et al. 2018; Lossau et al. 2019; Jung et al. 2020). For example, Lossau et al. (2019) developed a coronary motion forward artifact model for CT data (CoMoFACT) that is able to generate input and label data by introducing simulated noise to clinical CCTA images. Such a model learns the synthetic MVF and classifies/de-noises image patches accordingly. They achieved a mean error of 1.12 with an accuracy of $93.3\% \pm 1.8\%$ in classification between motion-free and motion-perturbed images. However, the clinical validity was missing especially in lack of hand-labelled data by radiologists and training on synthetic noise. Most recently, GAN models were also investigated in multiple studies, especially while training with generated synthetic images (Ren et al. 2022; Zhang et al. 2023a). Here, a pixel-to-pixel (pix2pix) GAN structure for image-to-image conversion was applied to estimate motion relative to motion-free images. The authors reported high performance with PSNR of 26.1 dB and SSIM of 0.864, however, their model was tested on images of the mid-right coronary artery (mRCA) only without investigating it on other vessels during motion. Dual-zone cycle Wasserstein GAN models were designed to learn motion displacement in image patches after applying a gradient penalty to synthetic images (Deng et al. 2021, 2022). Similarly, the performance of PSNR of 24.96 dB was evaluated on the left circumflex artery only with the model exhibiting high memory demands. Several other studies have explored different GAN architectures to handle motion artifacts (Xie et al. 2020; Van Herten et al. 2024), mostly on using basic structures to learn motion patterns. Moreover, they were focused on 2D slices rather than 3D volumes representing the temporal impact of motion on the scan. In another study by Gong et al. (2024), attention and spatial transformer (ATOM) sub-networks were developed using attention gating and cross-phase spatiotemporal information to handle CCTA motion noise. Here, motion-correction in reconstructed images with 140 ms and 75 ms temporal resolutions was tested with patients having > 60 bpm. In this study, ATOM has shown better performance in learning motion noise compared to other baseline models, i.e., U-Nets and ResNet, with a PSNR of 19.4 dB and SSIM of 0.870. However, this PSNR value is considered low compared to other studies, particularly since the study utilized a smaller dataset based on a single scanner type.

The use of preprocessing, i.e., dynamic time warping-based centerline alignment and synthetic ground truth, with a CycleGAN was proposed in Saleem et al. (2024). Their approach reported a median DSC of 0.751 and a Hausdorff distance of 5.39 on 4D CCTA paired patches, compared to near-motion free references, highlighting the benefits of cycle consistency and patch alignment. As an alternative to GANs, ATOM Gong et al. (2024) combines attention and spatial transformers, demonstrating improvements in both phantom (SSIM of 0.95, PSNR of 22.5 dB) and patient images (SSIM of 0.88 and PSNR of 32 dB) over uncorrected data, acknowledged through higher quality scores from radiologists. Overall, GAN approaches result in sharper edge delineations, compared to traditional CNNs, however, deep learning models are affected by limitations such as the diversity of the training datasets, and the focus on specific coronary vessels.

2.5 Other types of noise

2.5.1 Photon starvation artifacts

Photon starvation artifacts are a major cause of streaks in CT imaging (Mori et al. 2013; Hao et al. 2020), which can negatively affect the diagnosis of coronary artery disease in CCTA Nieman et al. (2024). This takes place when an insufficient number of photons reach the detector elements during low-dose scanning or thin-slice CT imaging, causing the X-ray beams to get severely attenuated and, thus, missing essential projection data (Zeng 2022b). This phenomenon often occurs near metal implants, metals with a high atomic number, or near-bone areas, i.e., pelvis or lungs (Selles et al. 2023). Moreover, since recent CT imaging protocols suggest shorter scan time and thinner slices (Klingenbeck-Regn et al. 1999), fewer photons reach the detectors, and thus, this low-mA imaging protocol causes photon starvation artifacts to appear more frequently (Nakamura et al. 2023).

Several studies implemented different approaches to handle the image artifacts caused such phenomena. For example, weighted IR was proposed by Shi et al. (2011) to mitigate the noise within the reconstructed images due to poor photon statistics (Mori et al. 2013). Sinogram smoothing algorithms followed by analytical reconstruction (AR) were proposed in various studies La Riviere (2005), Wang et al. (2006) to reduce the streaking effect in low-dose images by optimizing a penalty-based objective function, i.e., weighted least-squares. However, these approaches may over-smooth edges of anatomical structures reducing their diagnostic value especially near high contrast areas. Recent trends in CT imaging implemented adaptive filtering Singh et al. (2013), Wang et al. (2015), wavelet transform Hao et al. (2020), Zavala-Mondragon et al. (2022), shift-variant filtering Zeng (2022b), and photon-counting (Byl et al. 2021; Marcus et al. 2024). However, the majority of these methods rely on hyperparameter optimization and require continuous fine-tuning to ensure their reliability under different scenarios.

Deep learning methods are the most recently applied techniques to adjust for photon starvation artifacts. A number of studies designed spectral U-Net models Bauer et al. (2021), Busi et al. (2022) and tensor neural network (TNN) Shi et al. (2020), Shi et al. (2024) to learn, segment, and de-noise images due to photon starvation, i.e., streaks. However, the authors reported mild blurring in the reconstructed images due to hardware limitations, when implementing these heavy computational models. Moreover, spectral models usually suffer from scalability and generalization of spectral embedding (Xu et al. 2021). Focusing on CCTA imaging, Law et al. (2022) proposed an approach based on tube current modulation (TCM) to reduce the impact of photon starvation noise on coronary images when patient thickness increases photon attenuation. It was reported that photon starvation appears in patients with body mass index (BMI) above 27 kg/m^2 only due to the implemented IR reconstruction.

2.5.2 Partial volume effect

Partial volume effect is a phenomenon that occurs due to the appearance of multiple organ tissues within a single voxel in CT images Rezaei et al. (2019). It causes severe blurring around the edges, which happens as a result of the blending of signal intensities coming from multiple tissue structures Triche et al. (2019). In fine structures, e.g., myocardium, it is often difficult to measure accurate uptake qualitatively and quantitatively, especially in

the presence of diseases, such as myocardial infarction, due to variations in wall thickness Erlandsson et al. (2012), which substantially degrades cardiac images because of the partial volume effect phenomenon. Eventually, these artifacts cause misinterpretation of the tissues and surrounding structures and lead to quantitative errors in the measurement of tissue density and signal intensity within the reconstructed images Sandfort et al. (2021).

Correcting this effect is an essential step to enhancing CT images and preventing degradation in image quality. Several studies investigated the use of point spread-function modeling embedded within the IR algorithm while incorporating time-of-flight information to enhance the screening of lung lesions (Armstrong et al. 2014; Vennart et al. 2017; Huang et al. 2021). For example, Merlin et al. (2015) and Rezaei et al. (2019) utilized Lucy–Richardson de-convolution with wavelet-based de-noising to limit the spread of partial volume artifacts within CT images. Despite efficiency in de-noising reaching a CNR of 49.1 with proper contrast recovery, this approach requires selection of initialization parameters, i.e., point spread function (PSF) kernels, which could affect the results. Other studies investigated the use of prior information about the anatomical structures within a maximum-a-posteriori (AMAP) reconstruction algorithm to prevent the partial volume effect (Shcherbinin and Celler 2011; Chan et al. 2013, 2016). However, as stated in the previous sections, using prior information depends on the quality of information or features, e.g., anatomical boundaries, which could also suffer from noise themselves.

Most recently, deep learning algorithms have also been investigated to handle this kind of artifact in CT imaging. Several studies have utilized deep CNN-based models Huang et al. (2018a), Spadea et al. (2019), Chen et al. (2024) and U-Net models Azimi et al. (2022), Wang et al. (2022a) to remove partial volume artifacts in CT imaging. Deep learning methods reached SSIM of 0.98 with a mean average error of 31 HU. However, they were not tested clinically in most studies. For example, Chen et al. (2024) utilized a B-spline motion model as the noise source with no real noisy data. Moreover, the temporal resolution was one-fifth of one gantry rotation. In addition, Azimi et al. (2022) reported that their U-Net model caused over-estimations when de-noising due to having a small receptive field and boundary estimation errors during the prediction at a given point. In CCTA, a coronary lumen segmentation algorithm was developed to determine the hemodynamic significance of coronary artery stenosis while reducing partial volume artifacts (Freiman et al. 2016, 2017). Despite improvement in maximal surface distance error (+39%) compared with other conventional approaches, the authors reported that their method required an initial assumption of a correct centerline of the coronary artery. Moreover, no expert validation was performed for clinical cases. A deep learning-based reconstruction algorithm was proposed to handle the reduction in radiation and contrast doses, while preserving the edges of coronary arteries in the presence of these blurring artifacts Li et al. (2023). Similarly, an SNR up to 31.33 dB was observed for the left anterior descending branch (LAD) compared to 16.96 dB using the conventional adaptive statistical iterative reconstruction (ASIR-V) approach. However, the algorithm was not tested under challenging conditions, i.e., in the presence of calcification or obesity.

2.5.3 Electronic noise-induced artifacts

Electronic noise is due to the different components in the CT machinery (Xie et al. 2017). In this context, the type of detectors plays a pivotal role in the amount and strength of noise

Table 4 Summary of the state-of-the-art studies on the use of deep learning in reducing motion artifacts

Study	Year	Modality	Organ	Image dimension	Dataset	Noise origin
Convolutional neural network (CNN)						
Lossau et al. (2019)	2019	CCTA	Heart	2D	17 subject scans (local dataset)	Synthetic
Jung et al. (2020)	2020	CCTA	Heart	2D	Phantom data (CAVAREV platform) 100 clinical CT scans	Synthetic
Zhi et al. (2021)	2021	CT	Lung	2D	Phantom data and 4 patient scans	Respiration
Thummerer et al. (2022)	2022	CT	Lung	2D	45 scans (Local dataset - 45 patients)	Synthetic
Generative adversarial network (GAN)						
Wang et al. (2020)	2020	CT	Brain	2D	6,291 scans (Local dataset)	Synthetic
Deng et al. (2022)	2022	CCTA	Heart	2D	13,640 images (Local dataset - 60 patients)	Cardiac
Zhang et al. (2023a)	2023	CCTA	Heart	2D	366 scans (Local dataset - 366 patients)	Cardiac
Transformer						
Gong et al. (2024)	2024	CCTA	Heart	2D	142 images (Local dataset - 71 patients)	Cardiac (140 ms and 75 ms temporal phases)
Study	Comparison reference	Network details	Input	Output	Noise/de-noise model	Performance Limitations
Convolutional neural network (CNN)						
Lossau et al. (2019)	Motion levels (Scale 1 to 10)	2D CNN (CoMo-FACT)	60x60 patches	Motion noise level	$d_c(t_{cc}, v)$ $= s \cdot m_c(v) \cdot \delta_c(t_{cc})$ $\delta_i = \frac{p_i}{\max_{j,k} p_j - p_k _2}$	Accuracy: 93.3% MAE: 1.12 - Model fine-tuning performed on synthetic noise - Small clinical dataset - No hand-labelled data from radiologists

Study	Comparison reference	Network details	Input	Output	Noise/de-noise model	Performance	Limitations
Jung et al. (2020)	Motion levels (Scale 1 to 5)	2D CNN	60x60 patches	Noise-free image	$\mathcal{L}_{content}(I_x, I_c)$ $= \frac{1}{2} I_x - I_c _2^2$ $\mathcal{L}_{style}(I_x, I_s)$ $= \frac{1}{2N_x^2 \times 2N_s^2} G_x - G_s _2^2$ $\mathcal{L}_{total}(I_x, I_c, I_s)$ $= \alpha \mathcal{L}_{content}(I_x, I_c) + \beta \mathcal{L}_{style}(I_x, I_s)$	DSC: 0.829 SSIM: 0.970 PSNR: 33.6	- Small clinical dataset - Synthetic ground-truth motion-less images
Zhi et al. (2021)	High-quality images	CycN-Net	512x512x210 images	Noise-free image	$min \mathcal{L}(\Phi)$ $= \frac{1}{N} \sum_i Y_t - f(S_t, X_{prior}, \Phi) _2^2$	SSIM: 0.849	- Small clinical dataset - Inclusion of prior image with low noise levels
Thummerer et al. (2022)	High-quality images (0%-phase)	2D CNN	512x512 images	Noise-free image	NA	DSC: 0.98 SSIM: 0.93 PSNR: 43.3	- Lengthy time reconstruction - No patient-specific motion correction label - No real ground-truth images
Generative adversarial network (GAN)							
Wang et al. (2020)	High-quality images	Progressive-growing GAN	NA	Noise-free image	NA	NA	- No quantitative evaluation was performed - Synthetic motion artifacts

Study	Comparison reference	Network details	Input	Output	Noise/de-noise model	Performance	Limitations
Deng et al. (2022)	High-quality images	WGAN-GP Generator: U-Net Discriminator: CNN	256x256 images	Noise-free image	$L_D(x_1, x_2) = E_{x_1}[D(G(x_1))] - E_{x_2}[D(X_2)]$ $+ \lambda E_{x_1}[(\ D(G(x_1))\ _2 - 1)^2]$ $L_G(x) = -E_x[D(G(x))]$	SSIM: 0.769 PSNR: 24.96	- left circumflex coronary artery motion artifacts was not corrected - Texture and edge of the heart was not considered - Heavy memory demands for training
Zhang et al. (2023a)	Snapshot Freeze Motion-free images	Pix2pix model Generator: U-Net Discriminator: CNN	256x256 images	Noise-free image	$\mathcal{L}_{cGAN}(G, D) = E_{x,y}[\log D(x, y)]$ $+ E_{x,z}[\log(1 - D(x, G(x, z)))]$ $\mathcal{L}_{L1}(G)$ $= E_{x,y,z}[\ y - G(x, z)\ _1]$	DSC: 5.38 SSIM: 0.864 PSNR: 26.1	- Model was tested on a single CT scanner type - Single artery evaluation only
Transformer Gong et al. (2024)	High-quality images	ATOM structure	80x80 patches	Noise-free image	$L = \frac{1}{M}$ $\sum (L_{MSE} + L_{IGC} + \lambda L_{Feat})$ $L_{MSE} = \ f_{ATOM} - f_{label}\ _2^2$ L_{IGC} $= \frac{1}{\mathcal{P}(f_{ATOM}, f_{label}) + \xi}$ $L_{Feat} = \phi_l(f_{ATOM}) - \phi_l(f_{label})\ _2^2$	DSC: 0.88 SSIM: 0.870 PSNR: 19.4	- Small clinical dataset - Single scanner type - No diagnostic tasks were included in the analysis

during the reconstruction. For example, detectors with analog electronic circuits are a major cause of noise, when the signals are converted to digital form Duan et al. (2013). If digital circuits are used, the noise is mitigated. In modern CT machines, the effect of electronic noise is negligible due to the current slice and dose selection protocols (Shefer et al. 2013; Kumari and Goswami 2023). However, most electronic noise appears in low-dose imaging or when the screened patient is obese (Duan et al. 2013). In addition, CT machines with traditional detector designs consisting of external boards with analog electrical current that transfers between photodiodes and radiation-sensitive materials, i.e., cadmium tungstate, often suffer from noise during image reconstruction. On the other hand, integrated electronics in modern commercial CT systems include photodiodes, pre-amplifiers, and analog-to-digital converters, all in the same chip for analog connections (Boedeker et al. 2007). Another source of electronic noise is the thermal agitation in CT electronic circuits. This noise originates due to the transmission of current through the resistor, i.e., agitation of electrons inside conductors (Goyal et al. 2018). It has a Gaussian nature and mostly affects fluoroscopic radiography, e.g., digital subtraction angiography (DSA), owing to the need for increased photon receptor sensitivity (Dance et al. 2014).

3 Evaluation of de-noising performance

Performance of different de-noising techniques is usually evaluated with a variety of metrics. These metrics often relate the original noise-free images, i.e., ground-truth, with the de-noised ones using different processing algorithms (Lei et al. 2023).

The most common evaluation metric is the peak signal-to-noise ratio (PSNR); which is represented as the ratio of the maximum signal power in the original image to the power of the corrupting noise in the noisy image (Rajagopal et al. 2024). It represents the numerical difference between two images without considering structural information. PSNR can be calculated as follows:

$$PSNR(I_d, I_o) = 10 \log_{10} \frac{\max(I_o)^2}{MSE(I_d, I_o)}, \quad (24)$$

where I_d and I_o are the de-noised and original CT images, respectively.

On the other hand, a commonly used metric is the structural similarity index measure (SSIM), which is preferred when the evaluation is based on the quality preservation in the de-noised image (Mudeng et al. 2022). It is an overall representation of the linear correlation, mean luminances, and consistent contrasts between two images. SSIM can be calculated as follows:

$$SSIM(I_d, I_o) = \frac{(2\mu_{I_d}\mu_{I_o} + C_1)(2\sigma_{I_d I_o} + C_2)}{(\mu_{I_d}^2\mu_{I_o}^2 + C_1)(\sigma_{I_d}^2\sigma_{I_o}^2 + C_2)}, \quad (25)$$

where μ is the mean, σ is the variance, and C_1 and C_2 are constant values.

Other commonly used metrics are the mean square error (MSE), root mean square error (RMSE), and the contrast-to-noise ratio (CNR). The former two error metrics, are calculated pixel-wise, through comparison of the two images and reflect differences according

to the Euclidean distance metric (Chhabra et al. 2013). The latter represents the contrast between the average image values in a ROI relative to the background, which can be compared between the de-noised and original image (Alsleem and Almohiy 2020). In general, the higher the CNR, the less noise is present in the image and, thus, the higher the overall quality.

From a clinical perspective, evaluation of image quality is based on the level of useful information, with regards to the imaged objects, i.e., coronary arteries, etc. For example, calcification volume score, coverage score, and mass score are often used to evaluate the effect of de-blooming algorithms relative to the ground-truth value, i.e., Agatston score (Shabestari 2013).

4 Recent advances in CT technology

CT machinery has continuously developed and improved during the last decade to boost the quality of reconstructed images, maximize the safety of usage, and increase its functional capabilities (Lell and Kachelrieß 2020). Advances in CT hardware led to enhancements in the image reconstruction process, led by the US Food & Drug Association (FDA) to prevent unnecessary radiation exposure and improve CT imaging protocols (McLeavy et al. 2021). These advances included increasing gantry rotation speed, adding more rows to detectors, boosting X-ray tubes, and adding technological enhancements such as photon-counting detectors, multi-energy capabilities, and artificial intelligence-assisted reconstruction algorithms (Hsieh et al. 2020).

4.1 Increasing rotational speed and number of slices

The development of hardware is ongoing and nowadays, gantry rotational speed has reached 250 msec allowing an isotropic spatial resolution of 0.25–0.5 mm (Lell and Kachelrieß 2020). Such advances enable scanning the entire torso in 1 s using the modern high-pitch helical CT scanner with less noise (Hayes et al. 2021). In addition, higher speed allows for providing more images, i.e., slices, within a short time and thus, safer use of the X-ray radiation with less exposure times for patients (McLeavy et al. 2021). For example, modern technology uses sophisticated X-ray tubes with the anode directly in contact with the cooling oil, which is different from previous technology, where cooling was achieved through heat radiation (Gulati et al. 2023). This supports increasing power up to 120 kW with good rates for heat dissipation of over 7 MHU/min. This is an important development for cardiac imaging applications and CCTA, which require high-resolution and a larger number of slices, while the heart is showing less movement (Hagen et al. 2023).

4.2 Enhanced resolution and image quality

Recent advances in detector technology allow for enhancing the overall temporal and spatial resolution of CT imaging. Detectors with smaller elements and detectors with photon counting technology, i.e., single-photon detector (SPD), have significantly improved resolution (Ferda et al. 2021; Graafen et al. 2022). For example, Toshiba introduced Aquilion One (Patel et al. 2019) with a detector array of 16 cm width, and General Electric produced

Revolution CT (Gauntt 2019) with similar capabilities, both able to handle cone-beam artifacts, heel effect, and other spatial resolution noise, commonly shown with larger cone angles (Li et al. 2012). In the modern Canon machines' new Aquilion Precision system, an ultra-high resolution (UHR) detector of pixel size 0.25×0.25 ensures geometrical efficiency in scanning with a much smaller focal spot (Lell and Kachelrieß 2023). The photon counting detector mode is now integrated to allow reading sub-pixels and macro-pixels, providing higher spatial resolution with less noise (McCollough et al. 2023). Today's CT systems have detectors of 192–320 rows in the z direction with a maximum isocenter width of 16 cm allowing for static and dynamic imaging (Kang 2019). In cardiac CT imaging, the CardioGraph system by General Electric has two tubes aligned on the same x-y position with one detector array, which reduces cone-beam CT artifacts and enhances image quality (Chaikriangkrai et al. 2014).

4.3 Dose reduction and safer use

Several considerations were followed in modern CT machinery to handle noise reduction, while maintaining high image resolution. Tube current modulation (TCM) led to 15% to 50% dose reduction, especially when it became automatically controlled in recent developments. For example, 3D TCM control provided high image quality for different patients size information derived from the topograms, i.e., increasing the volume of CT dose index (CTDI) and decreasing it for large and small patients, respectively (Saltybaeva et al. 2018). Low-kV scanning, i.e., 70–100 kV, is now considered as the new standard; especially as 0.2 mSv for chest CT and sub-mSv for CCTA ensuring less radiation (Lell and Kachelrieß 2020). Modern CT technology incorporates automatic selection of tube voltage and current according to patient-specific attenuation and other planned examination types (Papadakis and Damilakis 2019). More importantly, X-ray tubes with a high current supply and less voltage ensure less dosage. Pre-filtration is also in use in several protocols to block undesired radiation, prior to reaching the patient while ensuring high image quality (Weis et al. 2017).

4.4 AI-enabled image reconstruction

Commercial deep learning is continuously growing in the development of advanced CT machinery that uses built-in reconstruction algorithms, contrary to the conventional IR and FBP methods. For example, most recently, Canon has developed their novel deep learning reconstruction method, advanced intelligent clear-IQ engine (AiCE), which requires low radiation dose and ensures high image quality and fast image generation speed (Ichikawa et al. 2021). TrueFidelity by General Electric, Precise Image by Philips Healthcare, PixelShine by AlgoMedica, and ClariCTAI by ClariPi were also introduced through the support of AI-driven methods (Koetzier et al. 2023). These algorithms are trained on low-dose CT images reconstructed using FBP and high-dose images reconstructed using the conventional IR method; however, they learn through neural networks to restore noisy images (Ziabari et al. 2018). These commercial CT algorithms were approved by the US FDA (Koetzier et al. 2023) providing enhancements in image quality. In the clinical setting, this technology improved availability and affordability due to the advancements in supercomputer levels and computational power (McLeavy et al. 2021).

5 Discussion and future directions

This review summarized recent advances in deep learning algorithms for correcting photometric artifacts in CT imaging with a focus on CCTA. The rapid development of data-driven, artificial intelligence-assisted tools has enabled advances in de-noising and photometric correction of CT images, which plays a pivotal role in clinical applications, i.e., assessment of coronary arteries and stenosis grading based on CCTA. The current state-of-the-art focuses on implementing simple, yet effective, algorithms that can learn noisy patterns within images from varying datasets, e.g., phantoms, synthetic, and clinical, while reducing the loss of information compared with conventional approaches. This supports improvements towards reliable decision making in clinical applications.

5.1 Technical aspects

To put this in perspective, from a methodological standpoint, deep learning architectures have been effective in identifying noisy patterns within the image, which cause distortions to the actual objects over the edges or narrow regions, i.e., coronary vessels. Upon learning these complex and variable patterns from large image datasets, neural networks such as U-Nets, GANs, or transformers (You et al. 2022a, b, 2024) tend to segment or generate, respectively, images with fewer artifacts. In other approaches, the network predicts the noisy mathematical and spatial patterns, which are to extracted afterwards from the noisy image. Compared to conventional mathematical techniques, deep learning has a key advantage in relating image priors and non-linear noise features to prevent loss of information, while performing de-noising.

In this review, we systematically evaluate the properties of deep learning models with respect to performance, computational efficiency, training data requirements and generalizability. A cross-cutting analysis of quantitative performance, reported in Tables I-IV, reveals interesting patterns. Multiple deep learning architectures were utilized in artifact correction across studies, including CNN, U-Net, and GAN, each poses advantages and challenges to be addressed. CNN are commonly used owing to their simplicity, computational efficiency, and high performance in extracting local spatial features. They are more suited for tasks involving denoising local artifacts, i.e., blooming or beam hardening. In global denoising tasks such as low-dose Gaussian noise or motion artifacts, they often struggle limiting their effectiveness in correcting larger anatomical regions. U-Nets, models that were preferable for image segmentation tasks, show high performance in tasks requiring detailed spatial resolution, such as correcting structured Gaussian noise or segmentation-related artifacts. They were less utilized in motion artifact correction across the covered studies. This is due to having skip connections that allow feature reuse between encoding and decoding layers. However, they are sensitive to data imbalance and may require large, annotated data.

In the context of low-dose denoising, GAN architectures outperform traditional CNNs in terms of SSIM, however, specialized architectures, e.g., ATOM networks (Gong et al. 2024) demonstrate superior performance in the case of motion artifacts. GANs perform well in most types of noise due to their ability of learning the complex and spatially varying artifacts, i.e., blur or blooming, by generating perceptually plausible corrections. If not carefully optimized and constrained, these models may be challenging to train or produce realistic generated outputs. Generative models were more likely to be used in low-dose denoising

and motion artifact correction including diffusion models. Diffusion architectures, despite their promising performance in low-dose CT denoising, may pose challenges when it comes to clinical integration due to their parameter size and processing times (Liu et al. 2023). Due to their generative nature, GANs tend to perform better when trained with smaller datasets (Sun and Ng 2022a), compared to U-Net-based architectures, which typically require a large number of training data. Some of the commercial products, e.g., ClariPi and AlgoMedica, are vendor-neutral demonstrating more cross-platform robustness than vendor-dedicated solutions which are trained on proprietary datasets.

Most recently, diffusion models in low-dose CT de-noising applications have shown promising performance, compared with other networks. These models have a strength in learning with or without ground-truth label images, i.e., high-dose, or pure images, owing to the functionality of creating forward and backward paths of different noise steps and learning how to de-noise them on their own. While such architectures were investigated for low-dose correction, they could also be applied in other types of noise, e.g., blooming artifacts in CCTA. Diffusion models have emerged as a powerful generative paradigm for CCTA denoising, resulting to improvements in perceptual quality and visible artifact reduction. However, they are several challenges related to their use in medical imaging applications. For instance, the stochastic nature of the sampling process may lead to the generation of structures which were not present in the original dataset, affecting data fidelity and ultimately, impacting on diagnostic accuracy. Moreover, these models are designed to process Gaussian noise as input, which differs in nature from the complex noise processes present in CCTA, e.g., blooming, beam hardening, etc. These inherent model characteristics could affect accurate visualization of coronary vessels, calcifications and soft tissue, suggesting that further research is required to focus on improving data fidelity rather than simply focusing on image synthesis. Another consideration is that diffusion models are computationally complex, often requiring significant inference time and GPU resources. As a result, they still require adjustment to fit into time-sensitive clinical applications. More recently, studies have introduced the use of different approaches to significantly reduce memory demands and time by using selective fine-tuning (Wang et al. 2024a), patch diffusion (Bieder et al. 2024), and Swin transformers (Shi et al. 2025). Last but not least, further research is required to validate their performance on large and heterogeneous medical imaging datasets. In this regard, studies should be focused on describing their clinical feasibility, robustness across diverse patient populations, and interpretability in different diagnostic scenarios.

However, it remains a challenge to build these models with large multi-center or multi-country imaging datasets to meet generalization and robustness requirements. To address this, future research should focus on the use of data augmentation techniques, transfer learning, and domain adaptation methods. In CCTA, due to the limited availability of datasets and/or labeled ground truth for the noise component, deep learning models with built-in diverse and representative data could facilitate the direct implementation of these networks in clinical settings without the loss of important information. This should be taken into consideration for CCTA, especially due to the variable imaging protocols, patient characteristics, and noise patterns, where data curation and learning from wider and larger cohorts is a necessity. The majority of current deep learning models for artifact correction in CCTA are trained on homogenous datasets from single centers or vendors, often with retrospective and protocol-specific acquisition parameters. This limits generalizability and raises concerns regarding model robustness when applied to external populations or varied scanning

conditions. Studies have shown that domain shift between datasets can lead to performance deterioration, especially in image regions affected by pathology or noise (Sheller et al. 2020; Rieke et al. 2020). To mitigate this, external validation across multi-center datasets, as well as the adoption of federated learning frameworks, are being explored to enable model training across institutions, while preserving patient privacy. However, there is a notable lack of peer-reviewed clinical trials validating these models prospectively in heterogeneous patient cohorts. Acknowledging this limitation is critical, as it sets the stage for next-step translational research.

5.2 Clinical relevance

From a clinical perspective, proper de-noising of CT image artifacts with the help of deep learning can lead to enhanced visualization of small coronary vessels, quantification of stenosis, and assessment of calcified plaques. This, in turn, contributes to the early detection of coronary diseases, such as atherosclerosis, and leads to providing informed treatment plans and prognostic predictions. Implementation of clinical artificial intelligence-based algorithms within vendor CT machinery has been recently implemented in clinical workflow. To date, three vendors have introduced deep learning within their reconstruction mechanisms to reduce artifacts, e.g., Canon AiCE, GE TrueFidelity, and Philips Precise Image. These CT scanner manufacturers rely completely on deep learning reconstruction without any conventional algorithms, i.e., FBP or IR. Despite these advancements, fully integrating deep learning algorithms into routine clinical practice requires further research, as current implementations remain limited. Challenges such as computational resource constraints, including memory demands and processing times, pose barriers to their widespread adoption in fast-paced clinical environments, where timely and robust imaging is crucial.

These commercial tools leverage deep learning to improve SNR and tissue definition and reduce the overall artifacts in clinical settings. Despite sharing the same objectives, they vary significantly in terms of network architecture and type of used data. TrueFidelity utilizes supervised learning trained on data from low- and high-quality images. Its performance was optimized to fit with GE systems and may not be easily transferable to other platforms. AiCE benefits from a hybrid model that merges physics about imaged objects with data-driven approaches. It restores low-quality CT images without the need for multiple forward projected iteration cycles. Precise Image is specifically designed for Philips iDose and IMR reconstruction systems ensuring a balance between noise suppression and spatial resolution. It mostly performs well on images generated using filtered back projection, however, it ensures noise reduction capabilities comparable to advanced iterative reconstruction techniques. Moreover, commercial models have limited cross-platform generalizability. Each tool is developed specifically to the vendor-specific hardware and software characteristics. Deep learning models perform well on images from within the training dataset distribution, however, they suffer from poor generalization in the case of distribution shifts, introduced, for instance, by the use of different scanners and acquisition protocols. Another challenge is the proprietary datasets used in the training of vendor-specific DL methods, which create biases towards the characteristics present in the specific datasets, e.g., artifacts and reconstruction properties. Moreover, these models are trained to operate on state-of-the-art vendor specific CT scanners, and by and large demonstrate performance limitations on older equipment or system solutions from other manufacturers. As a result,

multi-center studies, longitudinal patient tracking, and AI algorithm testing may be difficult if multiple vendor-specific machinery are utilized. Further research is needed on developing vendor-independent denoising solutions for post-processing of multi-vendor platform CCTAs. A potential approach is to explore the capabilities of transfer learning for domain adaptation to achieve high performance across the variety of CT equipment and acquisition protocols. Ultimately, these issues raise the importance of bringing together the vendor and researcher communities with regulatory agencies to consider industry-wide standardization to ensure superior performance across platforms.

The majority of current deep learning models for artifact correction in CCTA are trained on homogenous datasets from single centers or vendors, often with retrospective and protocol-specific acquisition parameters. This limits generalizability and raises concerns regarding model robustness when applied to external populations or varied scanning conditions. Studies have shown that domain shift between datasets can lead to performance deterioration, especially in image regions affected by pathology or noise. To mitigate this, external validation across multi-center datasets, as well as the adoption of federated learning frameworks, are being explored to enable model training across institutions while preserving patient privacy (Sheller et al. 2020; Rieke et al. 2020). However, there is a notable lack of peer-reviewed clinical trials validating these models prospectively in heterogeneous patient cohorts. Acknowledging this limitation is critical, as it sets the stage for next-step translational research.

In clinical settings, accuracy, reliability, and early detection of CAD are essential for better diagnostics and treatment outcomes. High-risk plaques, including non-calcified and low-attenuation plaques, are often missed or mischaracterized due to motion artifacts, image noise, or suboptimal contrast. False positives or negatives due to current de-noising and reconstruction approaches reduce the accuracy of diagnosis and limit confidence assessment. Reducing radiation dose is another important aspect of clinical review, which novel methods need to address, especially for vulnerable groups such as people with diabetes. Improving de-noising has direct impact on the quality of image reconstruction supporting earlier diagnosis, individualized treatment planning, and better risk stratification, and potentially resulting to reduced incidence of acute cardiac events and improved long-term cardiovascular outcomes. In detail, CCTA is a first-line imaging modality for evaluating patients with suspected CAD, with high negative predictive value and growing use in risk stratification. However, diagnostic accuracy is directly affected by image quality, which can be compromised by motion, noise, or calcification-related artifacts, especially in patients with high heart rates, arrhythmias, or heavy coronary calcification (Knuuti et al. 2020; Haag et al. 2025). Artifacts may obscure non-calcified plaques, distort lumen boundaries, or mimic stenoses, leading to equivocal or false interpretations. Image quality limitations have also been associated with non-diagnostic segments, which delay care and often require downstream testing (Yi et al. 2021). Artifact correction and denoising in CCTA address critical limitations that impair diagnostic accuracy, particularly in patients suffering with tachycardia and pronounced calcific buildup, or suboptimal imaging conditions. By improving lumen visualization, plaque characterization, and image interpretability, these methods enhance diagnostic confidence, reduce unnecessary downstream testing, and support more precise clinical decision-making.

Although PSNR and SSIM are commonly reported in the image processing literature, they fail to reflect diagnostic performance in clinically meaningful terms. Improvements in

CCTA image quality through deep learning-based denoising have been shown to increase segment interpretability and reduce rates of non-diagnostic findings in challenging cases. For example, a prospective study evaluating a vendor-agnostic deep learning reconstruction algorithm reported improved signal-to-noise and contrast-to-noise ratios, with increased diagnostic confidence among experienced readers (Takafuji et al. 2022). Moreover, refined image quality can improve the reproducibility of coronary artery calcium (CAC) scoring, facilitate more accurate quantification of luminal narrowing, and enhance the reliability of CT-derived assessment of fractional flow reserve; particularly in patients with borderline lesions or diffuse atherosclerosis. These downstream impacts have the potential to alter clinical management decisions and reduce unnecessary invasive angiography.

While PSNR and SSIM offer reproducible assessments of image similarity, they are not designed to measure clinical utility (Pryde et al. 2025). Their value is diminished in scenarios where perceptual or diagnostic accuracy is more relevant than pixel-level fidelity. For clinical translation, evaluation should incorporate task-based metrics such as reader-based diagnostic accuracy, interobserver variability, or ROC curves in lesion detection studies Dendumrongsup et al. (2014). A number of recent studies have adopted this dual-metric framework, comparing technical image quality parameters with diagnostic correctness in blinded radiologist evaluations. Including such measures can better align model performance with actual clinical outcomes, and we suggest this as a future direction for both benchmarking and validation.

5.3 Advantages and disadvantages

Current cardiac imaging reconstruction and denoising techniques for CAD are limited by their resolution, particularly in managing artifacts, motion-induced blurring, and poor signal-to-noise ratios in patients with persistent arrhythmias or elevated heart rates. Conventional methods, such as Gaussian smoothing and wavelet-based filters, often remove both noise and clinically relevant features. This reduction in detail may impair the ability to identify non-calcified or mixed plaques, which usually exhibit low contrast and are therefore frequently underestimated. As a result, there is a substantial risk of misdiagnosis, which increases the likelihood of heart attacks due to plaque rupture (Chapman et al. 2025). Calcified plaques may lead to beam hardening and blooming artifacts, which reduces the accuracy of lumen assessment. Furthermore, positive remodeling and low-attenuation plaques, which are critical indicators of vulnerable lesions, are often misclassified due to suboptimal plaque-to-tissue contrast and inconsistencies in reconstruction protocols. Moreover, automatic segmentation of coronary arteries, quantification of stenosis severity, and classification of plaque composition, all currently rely on time-consuming and operator-dependent manual interpretation. However, deep learning approaches, when trained appropriately, can utilize multi-view or temporal imaging data for better precision, faster processing speed, and improved consistency in plaque detection (Kang et al. 2017; Shrivastava et al. 2025).

Denoising methods, such as wavelet transform-based algorithms, provide multi-resolution information, and suppress noise at different scales. However, they depend on the choice of wavelet functions and may introduce additional noise (Lin et al. 2024). Non-local means filtering reduces noise without significant blurring, especially with low dose CT scans, but is computationally expensive and not as accurate when mixed plaques are present (Sadia et al. 2024). Deep learning-based denoising provides advantages as it learns patterns from large datasets and is adaptable to different imaging modalities. However, there is a disadvantage

in the use of deep learning models, e.g., CNN and U-net, as they require large, accurately annotated datasets for training purposes, and their performance may deteriorate when dealing with imbalanced datasets (Chen et al. 2017). Image reconstruction methods for CT and MRI include filtered back-projection, which is computationally efficient yet sensitive to noise and motion artifacts. MBIR, ASIR, and deep learning-based algorithms have further improved reconstruction quality (Shameer et al. 2018).

It is worth noting that most applications in CCTA motion artifact correction have predominantly focused on 2D deep learning models, particularly 2D CNNs. While these provide high accuracy with efficient computational usage, they inherently denoise each slice independently, thereby neglecting crucial temporal and spatial continuities between adjacent slices. Given the dynamic nature of cardiac anatomy and motion, 2D models trained solely on single slices may fail to capture inter-slice dependencies and motion patterns that span across volumes (Pandey and Rathore 2025). Therefore, there is a compelling need for future research in exploring 3D models or spatiotemporal architectures, e.g., 3D U-Nets or convolutional-recurrent hybrids, which are potentially able to leverage volumetric and temporal coherence to enhance correction performance. 3D convolutions were used in the approach by Green et al. (2018) as a means of enforcing inter-slice continuity in the 3D CNN. In Kobayashi et al. (2023), temporal information across different cardiac phases was incorporated within the proposed 4D noise reduction model. A hybrid approach, i.e., using temporal information without the use of 3D convolutions, was proposed by Gong et al. (2024), where an attention gate and spatial transformer (ATOM) system demonstrated strong performance in handling CCTA motion noise.

Beyond motion artifacts, the detection of subtle vascular abnormalities, such as coronary stenoses or early-stage plaque formations, requires maintaining 3D anatomical consistency (Kamiya et al. 2025). These features may span multiple slices and thus, can be misinterpreted when evaluated on a single slice. In Li et al. (2020d), enhanced structural coherence was achieved through a combination of 3D convolutions with self-attention blocks on 64x64x3 patches. A 3D UNet approach was proposed in the context of multiphase functional cardiac CT angiography, which resulted in excellent agreement with expert left ventricular segmentations, demonstrating clinically meaningful anatomical consistency (Sandfort et al. 2024). Improved denoising performance was achieved by leveraging inter-slice correlation in 3D CT scans in the 3D-DDnet system Kasparian et al. (2023), emphasizing the limitations of slice-by-slice processing. Moreover, the current literature primarily focuses on 2D evaluation metrics, e.g., PSNR and SSIM, which do not reflect 3D consistency. Future research should not only be designed to capture 3D knowledge about the volumetric structure but also employ loss functions that enforce coherence across the volume. Moreover, 3D anatomical details should be evaluated with metrics that assess structural continuity across slices, e.g., inter-slice continuity, and 3D structural information, such as volume overlap measures, thus supporting diagnostic confidence in vascular structure reconstruction.

Current state-of-the-art deep learning CCTA methods face challenges related to implementation and integration. Recent studies have shown that techniques such as diffusion models and 3D spatiotemporal networks require high-end GPUs with at least 24GB memory for clinical deployment, with inference times varying dramatically from 2–15 s per CCTA volume depending on model architecture (Ghekiere et al. 2017). The hierarchical ConvLSTM approach achieves faster computation than traditional U-Net, however, it requires hardware accelerators for real-time application. Clinical integration brings about additional

challenges, e.g., PACS integration, storage infrastructure for model configurations, and workflow disruption during model updates. Additionally, GANs are models that were often reported to have increased training instability and higher resource requirements due to multiple adversarial loss functions and multi-component pipelines (Fan et al. 2024). Vendor-specific implementations, e.g., AiCE and TrueFidelity operate at different computational efficiency levels, with studies showing processing times of 3-7 min for complete CCTA reconstruction versus 15-30 s for conventional methods. Future research should explore lightweight models, capable of delivering superior performance at reasonable computational costs, as memory and processing constraints remain substantial barriers to widespread implementation in time-sensitive clinical environments.

Lastly, research into deep learning models in CAD detection has highlighted their efficacy but have not been given regulatory approvals from relevant organizations, e.g., U.S. Food and Drug Administration (FDA), European Medicines Agency (EMA). Achieving such approvals requires extensive validation under real-world conditions to ensure safety, efficacy, and reproducibility. The path from algorithm development to clinical implementation is complex and constrained by regulatory, interpretability, and integration challenges. In the U.S., deep learning models that influence clinical decision-making fall under the FDA's Software as a Medical Device (SaMD) category, requiring evidence of analytical and clinical validity. In addition, there is currently a lack of explainable AI methods in this field, which will support overcoming the "black-box" outcomes currently obtained by deep learning models. This may be addressed by introducing saliency mapping, Grad-CAM, or attention mechanisms that visualize which image regions contributed to a given decision (Nannini et al. 2024; Ennab and Mcheick 2025). Therefore, beyond regulatory approval, model interpretability remains a central concern. For cardiac imaging applications, clinicians often require visual explanation tools to understand why a model has identified a region as pathological or artifact-laden. These tools enhance transparency but are rarely standardized, and their interpretive value can vary depending on the model architecture. Ultimately, prospective reader studies and regulatory-grade validations are necessary before these models can be reliably integrated into clinical workflows. Consistency across diverse populations, scanner types and imaging protocols are areas that need to be addressed in addition to long-term performance monitoring that provides information on accountable AI (Samant et al. 2025). To achieve integration of deep learning models into existing clinical workflows, there are technical challenges that need to be addressed, e.g., PACS, and user interface design to support effective interaction with clinicians. These need to be balanced with training strategies to build trust in AI recommendations.

6 Conclusion

The advancement of deep learning for artifact correction in CCTA has the potential to significantly improve cardiac imaging, however, its clinical adoption depends on rigorous external validation and testing of generalizability across diverse clinical settings. Although these algorithms outperform traditional correction methods, their effectiveness must be confirmed through comprehensive validation studies involving varied patient cohorts, imaging protocols, and scanner technologies. Differences in clinical conditions, patient anatomy, and image acquisition parameters present challenges that require robust testing to ensure consis-

tent performance in real-world environments. Future research should focus on establishing standardized validation protocols, including multi-center trials with diverse populations, to verify that these methods are accurate and adaptable across different clinical contexts. Techniques such as domain adaptation and transfer learning could also improve the generalizability of models, enabling them to perform reliably on data from varied sources without extensive retraining. Addressing these aspects is critical to achieving consistent, reproducible results and ensuring the safe integration of deep learning-based artifact correction in clinical practice. By prioritizing external validation and adaptability, this field can drive significant improvements in diagnostic precision and patient outcomes in cardiac imaging.

Acknowledgements This work was funded by Khalifa University, under grant RIG-2023-050.

Author contributions M.A. wrote the main manuscript text, P.L. thoroughly supervised and reviewed the study, E.A., H.F.J., and A.K. reviewed the medical aspects, and M.A. prepared figures and tables. All authors reviewed the manuscript.

Declarations

Conflict of interest The authors declare no Conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

- Abdelrahman KM, Chen MY, Dey AK, Virmani R, Finn AV, Khamis RY, Choi AD, Min JK, Williams MC, Buckler AJ et al (2020) Coronary computed tomography angiography from clinical uses to emerging technologies: Jacc state-of-the-art review. *J Am Coll Cardiol* 76(10):1226–1243
- Abdullah KA, McEntee MF, Reed W, Kench PL (2016) Radiation dose and diagnostic image quality associated with iterative reconstruction in coronary ct angiography: a systematic review. *J Med Imaging Radiat Oncol* 60(4):459–468
- Ackerman MJ (1998) The visible human project. *Proc IEEE* 86(3):504–511
- Agrawal H, Hietanen A, Särkkä S (2023) Deep learning based projection domain metal segmentation for metal artifact reduction in cone beam computed tomography. *IEEE Access* 11:100371–100382
- Ahmed O, Song Y (2018) A review of common beam hardening correction methods for industrial x-ray computed tomography. *Sains Malaysiana* 47(8):1883–1890
- Ahn CK, Jin H, Heo C, Kim JH (2019) “Combined low-dose simulation and deep learning for ct denoising: application of ultra-low-dose cardiac cta,” In: *Medical imaging 2019: Physics of medical imaging*, 10948, pp 1019–1024, SPIE
- Albrecht MH, Bickford MW, Schoepf UJ, Tesche C, De Santis D, Eid M, Jacobs BE, Duguay TM, Schmidt BT, Canstein C et al (2018) Beam-hardening in 70-kv coronary ct angiography: artifact reduction using an advanced post-processing algorithm. *Eur J Radiol* 101:111–117
- Alsleem HA, Almohiy HM (2020) The feasibility of contrast-to-noise ratio on measurements to evaluate ct image quality in terms of low-contrast detailed detectability. *Med Sci* 8(3):26

- Amirian M, Barco D, Herzig I, Schilling F-P (2024) “Artifact reduction in 3d and 4d cone-beam computed tomography images with deep learning-a review,” *IEEE Access*
- Andrew M, John H (2015) The challenge of coronary calcium on coronary computed tomographic angiography (ccta) scans: effect on interpretation and possible solutions. *Int J Cardiovasc Imaging* 31(Suppl 2):145–157
- Andreyev A, Yang F, Omlor L, Andrew M (2022) “Optimization of empirical beam hardening correction algorithm,” In: 7th International Conference on Image Formation in X-Ray Computed Tomography, **12304**, pp 701–706, SPIE
- Armstrong IS, Kelly MD, Williams HA, Matthews JC (2014) Impact of point spread function modelling and time of flight on fdg uptake measurements in lung lesions using alternative filtering strategies. *EJNMMI phys* 1:1–18
- Arunmuthu K, Ashish M, Saravanan T, Philip J, Rao B, Jayakumar T (2013) Simulation of beam hardening in x-ray tomography and its correction using linearisation and pre-filtering approaches. *Insight-Non-Destr Test Cond Monitor* 55(10):540–547
- Athanasiou LS, Fotiadis DI, Michalis LK (2017) Atherosclerotic plaque characterization methods based on coronary imaging. Academic Press, New York
- Ay MR, Mehranian A, Maleki A, Ghadiri H, Ghafarian P, Zaidi H (2013) Experimental assessment of the influence of beam hardening filters on image quality and patient dose in volumetric 64-slice x-ray ct scanners. *Physica Med* 29(3):249–260
- Azimi M-S, Kamali-Asl A, Ay M-R, Zeraatkar N, Arabi H (2022) “Deep learning-based partial volume correction in standard and low-dose pet-ct imaging,” Preprint at [arXiv:2207.02553](https://arxiv.org/abs/2207.02553)
- Barrett JF, Keat N (2004) Artifacts in ct: recognition and avoidance. *Radiographics* 24(6):1679–1691
- Bartykowszki A, Horcsik D, Kolossvary M, Jermendy A, Merkely B, Maurovich-Horvat P, et al (2015) “Image quality of coronary cta in heart transplant recipients,” In: European Congress of Radiology-ECR 2015
- Bauer DF, Ulrich C, Russ T, Golla A-K, Schad LR, Zöllner FG (2021) End-to-end deep learning ct image reconstruction for metal artifact reduction. *Appl Sci* 12(1):404
- Bayaraa T, Hyun CM, Jang TJ, Lee SM, Seo JK (2020) A two-stage approach for beam hardening artifact reduction in low-dose dental cbct. *Ieee Access* 8:225981–225994
- Belloir J-M, Lincelles J-B, Pelamatti A, Durnez C, Goiffon V, Virmontois C, Paillet P, Magnan P, Gilardx O (2017) Dark current blooming in pinned photodiode cmos image sensors. *IEEE Trans Electron Devices* 64(3):1161–1166
- Benz DC, Fuchs TA, Gräni C, Studer Bruengger AA, Clerc OF, Mikulicic F, Messerli M, Stehli J, Possner M, Pazhenkottil AP et al (2018) Head-to-head comparison of adaptive statistical and model-based iterative reconstruction algorithms for submillisievert coronary ct angiography. *Europ Heart J-Cardiovasc Imaging* 19(2):193–198
- Benz DC, Benetos G, Rampidis G, Von Felten E, Bakula A, Sustar A, Kudura K, Messerli M, Fuchs TA, Gebhard C et al (2020) Validation of deep-learning image reconstruction for coronary computed tomography angiography: impact on noise, image quality and diagnostic accuracy. *J Cardiovasc Comput Tomogr* 14(5):444–451
- Bhagalia R, Pack JD, Miller JV, Iatrou M (2012) Nonrigid registration-based coronary artery motion correction for cardiac computed tomography. *Med Phys* 39(7Part1):4245–4254
- Bieder F, Wolleb J, Durrer A, Sandkuehler R, Cattin PC (2024) “Memory-efficient 3d denoising diffusion models for medical image processing,” In: Medical Imaging with Deep Learning, pp 552–567, PMLR
- Boas FE, Fleischmann D et al (2012) Ct artifacts: causes and reduction techniques. *Imaging Med* 4(2):229–240
- Boedeker KL, Cooper VN, McNitt-Gray MF (2007) Application of the noise power spectrum in modern diagnostic mdct: part i. measurement of noise power spectra and noise equivalent quanta. *Phys Med Biol* 52(14):4027
- Brabant L, Pauwels E, Dierick M, Van Loo D, Boone M, Van Hoorebeke L (2012) A novel beam hardening correction method requiring no prior knowledge, incorporated in an iterative reconstruction algorithm. *Ndt E Int* 51:68–73
- Bucher AM, Wichmann JL, Schoepf UJ, Wolla CD, Canstein C, McQuiston AD, Krazinski AW, De Cecco CN, Meinel FG, Vogl TJ et al (2016) Quantitative evaluation of beam-hardening artefact correction in dual-energy ct myocardial perfusion imaging. *Eur Radiol* 26:3215–3222
- Busi M, Kehl C, Frisvad JR, Olsen UL (2022) Metal artifact reduction in spectral x-ray ct using spectral deep learning. *J Imaging* 8(3):77
- Byl A, Klein L, Sawall S, Heinze S, Schlemmer H-P, Kachelrieß M (2021) Photon-counting normalized metal artifact reduction (nmar) in diagnostic ct. *Med Phys* 48(7):3572–3582
- Carrascosa PM, Cury RC, Deviggiano A, Capunay C, Campisi R, de Munain ML, Vallejos J, Tajer C, Rodriguez-Granillo GA (2015) Comparison of myocardial perfusion evaluation with single versus dual-energy ct and effect of beam-hardening artifacts. *Acad Radiol* 22(5):591–599

- Catapano F, Lisi C, Savini G, Olivieri M, Figliozzi S, Caracciolo A, Monti L, Francone M (2023) Deep learning image reconstruction algorithm for ccta: Image quality assessment and clinical application. *J Comput Assist Tomograph*. 10–1097
- Chaikriangkrai K, Choi SY, Nabi F, Chang SM (2014) Important advances in technology and unique applications to cardiovascular computed tomography. *Methodist Debakey Cardiovasc J* 10(3):152
- Chan C, Liu H, Gbroshtein Y, Stacy MR, Sinusas AJ, Liu C (2013) “Simultaneous partial volume correction and noise regularization for cardiac spect/ct,” In: 2013 IEEE Nuclear Science Symposium and Medical Imaging Conference (2013 NSS/MIC), pp 1–6, IEEE
- Chan C, Liu H, Gbroshtein Y, Stacy MR, Sinusas AJ, Liu C (2016) Noise suppressed partial volume correction for cardiac spect/ct. *Med Phys* 43(9):5225–5239
- Chang S, Huber NR, Marsh JF, Koons EK, Gong H, Yu L, McCollough CH, Leng S (2023) Pie-net: Prior-information-enabled deep learning noise reduction for coronary ct angiography acquired with a photon counting detector ct. *Med Phys* 50(10):6283–6295
- Chapman AR, Taggart C, Boeddinghaus J, Mills NL, Fox KA (2025) Type 2 myocardial infarction: challenges in diagnosis and treatment. *Eur Heart J* 46(6):504–517
- Chee G, O’Connell D, Yang Y, Singhrao K, Low D, Lewis J (2019) Mcsart: an iterative model-based, motion-compensated sart algorithm for cbct reconstruction. *Phys Med Biol* 64(9):095013
- Chen F, Zhang L, Yu H, (2015) “External patch prior guided internal clustering for image denoising,” In: Proceedings of the IEEE international conference on computer vision, pp 603–611
- Chen S, Xi X, Li L, Luo L, Han Y, Wang J, Yan B (2016) “A filter design method for beam hardening correction in middle-energy x-ray computed tomography,” In: Eighth International Conference on Digital Image Processing (ICDIP 2016), **10033**, pp 1004–1009, SPIE
- Chen H, Zhang Y, Kalra MK, Lin F, Chen Y, Liao P, Zhou J, Wang G (2017) Low-dose ct with a residual encoder-decoder convolutional neural network. *IEEE Trans Med Imaging* 36(12):2524–2535
- Chen K, Zhang L-B, Liu J-S, Gao Y, Wu Z, Zhu H-C, Du C-P, Mai X-L, Yang C-F, Chen Y (2022) Robust restoration of low-dose cerebral perfusion ct images using nes-unet. *Nucl Sci Tech* 33(3):30
- Chen Z, Li Q, Wu D (2024) Estimate and compensate head motion in non-contrast head ct scans using partial angle reconstruction and deep learning. *Med Phys* 51(5):3309–3321
- Chhabra T, Dua G, Malhotra T (2013) Comparative analysis of methods to denoise ct scan images. *Int J Adv Res Elect Electron Instrument Eng* 2(7):3363–3369
- Choi J-H, Maier A, Keil A, Pal S, McWalter EJ, Beaupré GS, Gold GE, Fahrig R (2014) Fiducial marker-based correction for involuntary motion in weight-bearing c-arm ct scanning of knees. ii. experiment. *Med Phys* 41(6Part1):061902
- Cousins C, Miller D, Bernardi G, Rehani M, Schofield P, Vañó E, Einstein A, Geiger B, Heintz P, Padovani R et al (2013) Icrp publication 120: radiological protection in cardiology. *Ann ICRP* 42(1):1–125
- Dance D, Christofides S, Maidment A, McLean I, Ng K (2014) Diagnostic radiology physics. *Int Atom Energy Agency* 299:12–14
- de Knecht MC, Haugen M, Jensen AK, Linde JJ, Kühl JT, Hove JD, Kofoed KF (2019) Coronary plaque composition assessed by cardiac computed tomography using adaptive hounsfield unit thresholds. *Clin Imaging* 57:7–14
- Dendumrongsup T, Plumb AA, Halligan S, Fanshawe TR, Altman DG, Mallett S (2014) Multi-reader multi-case studies using the area under the receiver operator characteristic curve as a measure of diagnostic accuracy: systematic review with a focus on quality of data reporting. *PLoS ONE* 9(12):e116018
- Deng F, Tie C, Zeng Y, Shi Y, Wu H, Wu Y, Liang D, Liu X, Zheng H, Zhang X et al (2021) Correcting motion artifacts in coronary computed tomography angiography images using a dual-zone cycle generative adversarial network. *J Xray Sci Technol* 29(4):577–595
- Deng F, Wan Q, Zeng Y, Shi Y, Wu H, Wu Y, Xu W, Mok GS, Zhang X, Hu Z (2022) Image restoration of motion artifacts in cardiac arteries and vessels based on a generative adversarial network. *Quant Imaging Med Surg* 12(5):2755
- Dhou S, Hurwitz M, Mishra P, Cai W, Rottmann J, Li R, Williams C, Wagar M, Berbeco R, Ionascu D et al (2015) 3d fluoroscopic image estimation using patient-specific 4dcbct-based motion models. *Phys Med Biol* 60(9):3807
- Dhou S, Alkhodari M, Ionascu D, Williams C, Lewis JH (2022) Fluoroscopic 3d image generation from patient-specific pca motion models derived from 4d-cbct patient datasets: a feasibility study. *J Imaging* 8(2):17
- Do S, Karl WC, Liang Z, Kalra M, Brady TJ, Pien HH (2011) A decomposition-based ct reconstruction formulation for reducing blooming artifacts. *Phys Med Biol* 56(22):7109
- Dong G, Zhang C, Deng L, Zhu Y, Dai J, Song L, Meng R, Niu T, Liang X, Xie Y (2022) A deep unsupervised learning framework for the 4d cbct artifact correction. *Phys Med Biol* 67(5):055012
- Du W, Chen H, Wu Z, Sun H, Liao P, Zhang Y (2017) Stacked competitive networks for noise reduction in low-dose ct. *PLoS ONE* 12(12):e0190069

- Duan X, Wang J, Leng S, Schmidt B, Allmendinger T, Grant K, Flohr T, McCollough CH (2013) Electronic noise in ct detectors: impact on image noise and artifacts. *Am J Roentgenol* 201(4):W626–W632
- Eckert J, Renczes-Janetzko P, Schmidt M, Magedanz A, Voigtländer T, Schmermund A (2019) Coronary ct angiography (ccta) using third-generation dual-source ct for ruling out in-stent restenosis. *Clin Res Cardiol* 108:402–410
- Elss T, Nickisch H, Wissel T, Bippus R, Morlock M, Grass M (2018) “Motion estimation in coronary ct angiography images using convolutional neural networks,” In: *Medical Imaging with Deep Learning*
- Ennab M, Mcheick H (2025) Advancing ai interpretability in medical imaging: a comparative analysis of pixel-level interpretability and grad-cam models. *Mach Learn Knowled Extr* 7(1):12
- Erlandsson K, Buvat I, Pretorius PH, Thomas BA, Hutton BF (2012) A review of partial volume correction techniques for emission tomography and their applications in neurology, cardiology and oncology. *Phys Med Biol* 57(21):R119
- Esposito A, Colantoni C, De Cobelli F, Del Vecchio A, Palmisano A, Calandrino R, Del Maschio A (2013) Multidetector computed tomography for coronary stents imaging: high-voltage (140-kvp) prospective ecg-triggered versus standard-voltage (120-kvp) retrospective ecg-gated helical scanning. *J Comput Assist Tomogr* 37(3):395–401
- European Union (2014) “European commission radiation protection no. 180,” Medical radiation exposure of the European population
- Fan F, Shan H, Kalra MK, Singh R, Qian G, Getzin M, Teng Y, Hahn J, Wang G (2019) Quadratic autoencoder (q-ae) for low-dose ct denoising. *IEEE Trans Med Imaging* 39(6):2035–2050
- Fan Y, Qin T, Sun Q, Wang M, Liang B (2024) A review of factors affecting radiation dose and image quality in coronary cta performed with wide-detector ct. *Tomography* 10(11):1730–1743
- Feng Z, Li Z, Cai A, Li L, Yan B, Tong L (2020) “A preliminary study on projection denoising for low-dose ct imaging using modified dual-domain u-net,” In: *2020 3rd International Conference on Artificial Intelligence and Big Data (ICAIBD)*, pp 223–226, IEEE
- Ferda J, Vendiš T, Flohr T, Schmidt B, Henning A, Ulzheimer S, Pecen L, Ferdová E, Baxa J, Mírka H (2021) Computed tomography with a full fov photon-counting detector in a clinical setting, the first experience. *Eur J Radiol* 137:109614
- Freiman M, Lamash Y, Gilboa G, Nickisch H, Prevrhal S, Schmitt H, Vembar M, Goshen L (2016) “Automatic coronary lumen segmentation with partial volume modeling improves lesions’ hemodynamic significance assessment,” In: *Medical Imaging 2016: Image Processing*, **9784**, pp 15–22, SPIE
- Freiman M, Nickisch H, Prevrhal S, Schmitt H, Vembar M, Maurovich-Horvat P, Donnelly P, Goshen L (2017) Improving ccta-based lesions’ hemodynamic significance assessment by accounting for partial volume modeling in automatic coronary lumen segmentation. *Med Phys* 44(3):1040–1049
- Fryar CD, Chen T-C, Li X (2012) Prevalence of uncontrolled risk factors for cardiovascular disease: United States, 1999–2010, 103. US Department of Health and Human Services, Centers for Disease Control and
- Fujii K, Nomura K, Imai K, Muramatsu Y, Tsushima S, Ota H (2022) Evaluation of apparent noise on ct images using moving average filters. *J Digit Imaging* 35(1):77–85
- Gao Q, Li Z, Zhang J, Zhang Y, Shan H (2023) Corediff: contextual error-modulated generalized diffusion model for low-dose ct denoising and generalization. *IEEE Trans Med Imaging* 43(2):745–759
- Gauntt DM (2019) A suggested method for setting up gsi profiles on the ge revolution ct scanner. *J Appl Clin Med Phys* 20(12):169–179
- Geramifar P, Zafarghandi MS, Ghafarian P, Rahmim A, Ay MR (2013) Respiratory-induced errors in tumor quantification and delineation in ct attenuation-corrected pet images: effects of tumor size, tumor location, and respiratory trace: a simulation study using the 4d xcat phantom. *Mol Imag Biol* 15:655–665
- Ghafarzadeh M, Kejani MT, Asadi A, Karimi M (2024) “A single material beam hardening reduction approach for industrial x-ray ct images utilizing convolutional neural network,” In: *13th International Conference on Industrial Computed Tomography*, Wels, Austria, Feb, pp 6–9
- Ghani MU, Karl WC (2019) Fast enhanced ct metal artifact reduction using data domain deep learning. *IEEE Trans Comput Imaging* 6:181–193
- Ghekiere O, Salgado R, Buls N, Leiner T, Mancini I, Vanhoenacker P, Dendale P, Nchimi A (2017) Image quality in coronary ct angiography: challenges and technical solutions. *Br J Radiol* 90(1072):20160567
- Gjesteby L, Shan H, Yang Q, Xi Y, Claus B, Jin Y, De Man B, Wang G (2018) “Deep neural network for ct metal artifact reduction with a perceptual loss function,” In: *The fifth international conference on image formation in X-ray computed tomography*, pp 439–443
- Gjesteby L, Shan H, Yang Q, Xi Y, Jin Y, Giantsoudi D, Paganetti H, De Man B, Wang G (2019) A dual-stream deep convolutional network for reducing metal streak artifacts in ct images. *Phys Med Biol* 64(23):235003

- Gong H, Ahmed Z, Chang S, Koons EK, Thorne JE, Rajiah P, Foley TA, Fletcher JG, McCollough CH, Leng S (2024) Motion artifact correction in cardiac ct using cross-phase temporospatial information and synergistic attention gate and spatial transformer sub-networks. *Phys Med Biol* 69(3):035023
- Goo HW (2012) Ct radiation dose optimization and estimation: an update for radiologists. *Korean J Radiol* 13(1):1
- Goyal B, Agrawal S, Sohi B (2018) Noise issues prevailing in various types of medical images. *Biomed Pharmacol J* 11(3):1227
- Graafen D, Emrich T, Halfmann MC, Mildenerberger P, Düber C, Yang Y, Othman AE, O'Doherty J, Müller L, Kloeckner R (2022) Dose reduction and image quality in photon-counting detector high-resolution computed tomography of the chest: routine clinical data. *J Thorac Imaging* 37(5):315–322
- Grant KL, Flohr TG, Krauss B, Sedlmair M, Thomas C, Schmidt B (2014) Assessment of an advanced image-based technique to calculate virtual monoenergetic computed tomographic images from a dual-energy examination to improve contrast-to-noise ratio in examinations using iodinated contrast media. *Invest Radiol* 49(9):586–592
- Green M, Marom EM, Kiryati N, Konen E, Mayer A (2016) “Efficient low-dose ct denoising by locally-consistent non-local means (lc-nlm),” In: *Medical Image Computing and Computer-Assisted Intervention-MICCAI 2016: 19th International Conference, Athens, Greece, October 17–21, 2016, Proceedings, Part III* 19, pp 423–431, Springer
- Green M, Marom EM, Konen E, Kiryati N, Mayer A (2018) 3-d neural denoising for low-dose coronary ct angiography (ccta). *Comput Med Imaging Graph* 70:185–191
- Grosser OS, Ruf J, Kupitz D, Czuczvara D, Loewenthal D, Thormann M, Furth C, Ricke J, Denecke T, Pech M et al (2019) Iterative ct reconstruction in abdominal low-dose ct used for hybrid spect-ct applications: effect on image quality, image noise, detectability, and reader's confidence. *Acta Radiologica Open* 8(6):2058460119856266
- Gu S, McNamara JE, Mitra J, Gifford HC, Johnson K, Gennert MA, King MA (2010) Body deformation correction for spect imaging. *IEEE Trans Nucl Sci* 57(1):214–224
- Gu J, Yang TS, Ye JC, Yang DH (2021) Cyclegan denoising of extreme low-dose cardiac ct using wavelet-assisted noise disentanglement. *Med Image Anal* 74:102209
- Gulati P, Chidambaranathan N, Ahuja A, Anbarasu A, Mahajan A (2023) *Comprehensive Textbook of Clinical Radiology Volume I: Principles of Clinical Radiology, Multisystem Diseases & Head and Neck-E-book*. Elsevier Health Sciences
- Gunawan R, Tran Y, Zheng J, Nguyen H, Chai R (2022) Image recovery from synthetic noise artifacts in ct scans using modified u-net. *Sensors* 22(18):7031
- Guo M, Chee G, O'Connell D, Dhou S, Fu J, Singhrao K, Ionascu D, Ruan D, Lee P, Low DA et al (2019) Reconstruction of a high-quality volumetric image and a respiratory motion model from patient cbct projections. *Med Phys* 46(8):3627–3639
- Haag NP, Niehoff JH, Shahzadi I, Panknin C, Wiemer M, Kaese S, Gertz RJ, Pennig L, Platte OI, Surov A et al (2025) Improving image quality and diagnostic usability in photon-counting coronary ct angiography using a novel reconstruction algorithm. *Europ Radiol* 1–7
- Hagen F, Estler A, Hofmann J, Walder L, Faby S, Almarie B, Nikolaou K, Wrazidlo R, Horger M (2023) Reduced versus standard dose contrast volume for contrast-enhanced abdominal ct in overweight and obese patients using photon counting detector technology vs. second-generation dual-source energy integrating detector ct. *Eur J Radiol* 169:111153
- Han M, Shim H, Baek J (2021) Low-dose ct denoising via convolutional neural network with an observer loss function. *Med Phys* 48(10):5727–5742
- Hanna RD, Ketcham RA (2017) X-ray computed tomography of planetary materials: a primer and review of recent studies. *Geochemistry* 77(4):547–572
- Hao S, Liu J, Chen Y, Liu B, Wei C, Zhu J, Li B (2020) A wavelet transform-based photon starvation artifacts suppression algorithm in ct imaging. *Phys Med Biol* 65(23):235039
- Hashemi S, Paul NS, Beheshti S (2015) Adaptively tuned iterative low dose ct image denoising. *Comput Math Methods Med* 1:638568
- Hasinoff SW (2021) “Photon, poisson noise,” In: *Computer vision: a reference guide*, pp 980–982, Springer
- Hayes JW, Montoya J, Budde A, Zhang C, Li Y, Li K, Hsieh J, Chen G-H (2021) High pitch helical ct reconstruction. *IEEE Trans Med Imaging* 40(11):3077–3088
- He T, Qian X, Zhai R, Yang Z (2015) Computed tomography number measurement consistency under different beam hardening conditions: comparison between dual-energy spectral computed tomography and conventional computed tomography imaging in phantom experiment. *J Comput Assist Tomogr* 39(6):981–985
- Heinrich MP, Stille M, Buzug TM (2018) Residual u-net convolutional neural network architecture for low-dose ct denoising. *Curr Directions in Biomed Eng* 4(1):297–300
- Hermena S, Young M (2023) “Ct-scan image production procedures,” *StatPearls*

- Higaki T, Nakamura Y, Zhou J, Yu Z, Nemoto T, Tatsugami F, Awai K (2020) Deep learning reconstruction at ct: phantom study of the image characteristics. *Acad Radiol* 27(1):82–87
- Hofmann C, Knaup M, Kachelrieß M (2014) “Removing blooming artifacts with binarized deconvolution in cardiac ct,” In: *Medical Imaging 2014: Physics of Medical Imaging*, **9033**, pp. 153–162, SPIE
- Hong JH, Park E-A, Lee W, Ahn C, Kim J-H (2020) Incremental image noise reduction in coronary ct angiography using a deep learning-based technique with iterative reconstruction. *Korean J Radiol* 21(10):1165
- Hsieh SS, Leng S, Rajendran K, Tao S, McCollough CH (2020) Photon counting ct: clinical applications and future developments. *IEEE Trans Radiat Plasma Med Sci* 5(4):441–452
- Huang B, Chen Z, Wu P-M, Ye Y, Feng S-T, Wong C-YO, Zheng L, Liu Y, Wang T, Li Q et al (2018a) Fully automated delineation of gross tumor volume for head and neck cancer on pet-ct using deep learning: A dual-center study. *Contrast Med Mole Imaging* 2018(1):8923028
- Huang X, Zhang Y, Wang J (2018b) A biomechanical modeling-guided simultaneous motion estimation and image reconstruction technique (smear-bio) for 4d-cbct reconstruction. *Phys Med Biol* 63(4):045002
- Huang L, Jiang H, Li S, Bai Z, Zhang J (2020a) Two stage residual cnn for texture denoising and structure enhancement on low dose ct image. *Comput Methods Programs Biomed* 184:105115
- Huang X, Zhang Y, Chen L, Wang J (2020b) U-net-based deformation vector field estimation for motion-compensated 4d-cbct reconstruction. *Med Phys* 47(7):3000–3012
- Huang K, Feng Y, Liang W, Li L (2021) Impact of time of flight and point spread function on quantitative parameters of lung lesions in 18 f-fdg pet/ct. *BMC Med Imaging* 21:1–9
- Hunter AK, McDavid W (2012) Characterization and correction of cupping effect artefacts in cone beam ct. *Dentomaxillofacial Radiol* 41(3):217–223
- Ibrahim EA, Al Awar S, Balayah ZH, Hadjileontiadis LJ, Khandoker AH (2017) A comparative study on fetal heart rates estimated from fetal phonography and cardiotocography. *Front Physiol* 8:764
- Ichikawa Y, Kanii Y, Yamazaki A, Nagasawa N, Nagata M, Ishida M, Kitagawa K, Sakuma H (2021) Deep learning image reconstruction for improvement of image quality of abdominal computed tomography: comparison with hybrid iterative reconstruction. *Jpn J Radiol* 39:598–604
- Imai K, Ikeda M, Enchi Y, Niimi T (2009) Statistical characteristics of streak artifacts on ct images: relationship between streak artifacts and ma s values. *Med Phys* 36(2):492–499
- Immonen E, Wong J, Nieminen M, Kekkonen L, Roine S, Törnroos S, Lanca L, Guan F, Metsälä E (2022) The use of deep learning towards dose optimization in low-dose computed tomography: a scoping review. *Radiography* 28(1):208–214
- Infante T, Cavaliere C, Punzo B, Grimaldi V, Salvatore M, Napoli C (2021) Radiogenomics and artificial intelligence approaches applied to cardiac computed tomography angiography and cardiac magnetic resonance for precision medicine in coronary heart disease: a systematic review. *Circ Cardiovasc Imaging* 14(12):1133–1146
- Jeon P-H, Jeon S-H, Ko D, An G, Shim H, Otgonbaatar C, Son K, Kim D, Ko SM, Chung M-A (2023) Assessment of image quality of coronary ct angiography using deep learning-based ct reconstruction: phantom and patient studies. *Diagnostics* 13(11):1862
- Jeong H, Park T, Khang S, Koo K, Shin J, Kim KW, Lee J (2023) Non-rigid registration based on hierarchical deformation of coronary arteries in ccta images. *Biomed Eng Lett* 13(1):65–72
- Ji X, Gao D, Gan Y, Zhang Y, Xi Y, Quan G, Lu Z, Chen Y (2023) A deep-learning-based method for correction of bone-induced ct beam-hardening artifacts. *IEEE Trans Instrum Meas* 72:1–12
- Jiang Z, Zhang Z, Chang Y, Ge Y, Yin F-F, Ren L (2021) Enhancement of 4-d cone-beam computed tomography (4d-cbct) using a dual-encoder convolutional neural network (decnn). *IEEE Trans Radiat Plasma Med Sci* 6(2):222–230
- Jiang S, Sun Y, Xu S, Zhang Z, Wu Z (2024) Metal artifact correction in industrial ct images based on a dual-domain joint deep learning framework. *Appl Sci* 14(8):3261
- Jin P, Bouman CA, Sauer KD (2015) A model-based image reconstruction algorithm with simultaneous beam hardening correction for x-ray ct. *IEEE Trans Comput Imaging* 1(3):200–216
- Jing J, Xia W, Hou M, Chen H, Liu Y, Zhou J, Zhang Y (2022) Training low dose ct denoising network without high quality reference data. *Phys Med Biol* 67(8):084002
- Jung S, Lee S, Jeon B, Jang Y, Chang H-J (2018) “Deep learning based coronary artery motion artifact compensation using style-transfer synthesis in ct images,” In: *Simulation and Synthesis in Medical Imaging: Third International Workshop, SASHIMI 2018, Held in Conjunction with MICCAI 2018, Granada, Spain, September 16, 2018, Proceedings 3*, pp 100–110, Springer
- Jung S, Lee S, Jeon B, Jang Y, Chang H-J (2020) Deep learning cross-phase style transfer for motion artifact correction in coronary computed tomography angiography. *IEEE Access* 8:81849–81863
- Kalare K, Bajpai M, Sarkar S, Munshi P (2022) Deep neural network for beam hardening artifacts removal in image reconstruction. *Appl Intell* 52(6):6037–6056
- Kalisz K, Bueche K, Saboo SS, Abbasa S, Halliburton S, Rajiah P (2016) Artifacts at cardiac ct: physics and solutions. *Radiographics* 36(7):2064–2083

- Kamiya K, Nagatani Y, Matsubayashi J, Uemura R, Oki T, Matsubayashi Y, Terada S, Vigers P, Nakata S, Watanabe Y et al (2025) Usefulness of isosurface geometric measurement on volume-rendered images for quantitative measurements of complex cardiac anatomical features. *J Card Surg* 2025(1):5193639
- Kang E-J (2019) Clinical applications of wide-detector ct scanners for cardiothoracic imaging: an update. *Korean J Radiol* 20(12):1583
- Kang E, Min J, Ye JC (2017) A deep convolutional neural network using directional wavelets for low-dose x-ray ct reconstruction. *Med Phys* 44(10):e360–e375
- Kang E, Chang W, Yoo J, Ye JC (2018) Deep convolutional framelet denosing for low-dose ct via wavelet residual network. *IEEE Trans Med Imaging* 37(6):1358–1369
- Kasparian A, Cao G, Feng W-c (2023) “A 3d deep learning architecture for denoising low-dose ct scans,” In: *International Conference on Computational Advances in Bio and Medical Sciences*, pp. 94–106, Springer
- Kay FU, Canan A, Abbasa S (2020) Future directions in coronary ct angiography: Ct-fractional flow reserve, plaque vulnerability, and quantitative plaque assessment. *Korean Circul J* 50(3):185
- Ketcham RA, Hanna RD (2014) Beam hardening correction for x-ray computed tomography of heterogeneous natural materials. *Comput Geosci* 67:49–61
- Khaleghi G, Hosntalab M, Sadeghi M, Reiazi R, Mahdavi SR (2021) Metal artifact reduction in computed tomography images based on developed generative adversarial neural network. *Inform Med Unlocked* 24:100573
- Kim J, Nuyts J, Kyme A, Kuncic Z, Fulton R (2015) A rigid motion correction method for helical computed tomography (ct). *Phys Med Biol* 60(5):2047
- Kim B, Han M, Shim H, Baek J (2019) A performance comparison of convolutional neural network-based image denoising methods: The effect of loss functions on low-dose ct images. *Med Phys* 46(9):3906–3923
- Kim J, Kim J, Han G, Rim C, Jo H (2020) Low-dose ct image restoration using generative adversarial networks. *Inform Med Unlocked* 21:100468
- Kitagawa K, George RT, Arbab-Zadeh A, Lima JA, Lardo AC (2010) Characterization and correction of beam-hardening artifacts during dynamic volume ct assessment of myocardial perfusion. *Radiology* 256(1):111–118
- Klima O, Chromy A, Zemcik P, Spanel M, Kleparnik P (2016) A study on performace of levenberg-marquardt and cma-es optimization methods for atlas-based 2d/3d reconstruction. *IFAC-PapersOnLine* 49(25):121–126
- Klingenbeck-Regn K, Schaller S, Flohr T, Ohnesorge B, Kopp AF, Baum U (1999) Subsecond multi-slice computed tomography: basics and applications. *Eur J Radiol* 31(2):110–124
- Klug M, Shemesh J, Green M, Mayer A, Kerpel A, Konen E, Marom E (2022) A deep-learning method for the denoising of ultra-low dose chest ct in coronary artery calcium score evaluation. *Clin Radiol* 77(7):e509–e517
- Knuuti J, Wijns W, Saraste A, Capodanno D, Barbato E, Funck-Brentano C, Prescott E, Storey RF, Deaton C, Cuisset T et al (2020) 2019 esc guidelines for the diagnosis and management of chronic coronary syndromes: the task force for the diagnosis and management of chronic coronary syndromes of the european society of cardiology (esc). *Eur Heart J* 41(3):407–477
- Ko Y, Moon S, Baek J, Shim H (2021) Rigid and non-rigid motion artifact reduction in x-ray ct using attention module. *Med Image Anal* 67:101883
- Kobayashi T, Nishii T, Umehara K, Ota J, Ohta Y, Fukuda T, Ishida T (2023) Deep learning-based noise reduction for coronary ct angiography: using four-dimensional noise-reduction images as the ground truth. *Acta Radiol* 64(5):1831–1840
- Koetzier LR, Mastrodicasa D, Szczykutowicz TP, van der Werf NR, Wang AS, Sandfort V, van der Molen AJ, Fleischmann D, Willemink MJ (2023) Deep learning image reconstruction for ct: technical principles and clinical prospects. *Radiology* 306(3):e221257
- Kovan B (2024) The effects of kilovoltage (kv) and milliampere seconds (mas) values for spect attenuation correction: an anthropomorphic phantom study. *Appl Radiat Isot* 204:111118
- Kruk M, Noll D, Achenbach S, Mintz GS, Pręgowski J, Kaczmarek E, Kryczka K, Pracon R, Dzielińska Z, Śleszycka J et al (2014) Impact of coronary artery calcium characteristics on accuracy of ct angiography. *JACC Cardiovasc Imaging* 7(1):49–58
- Kulathilake K, Abdullah NA, Sabri A, Lai KW (2023) A review on deep learning approaches for low-dose computed tomography restoration. *Complex Intell Syst* 9(3):2713–2745
- Kumari K, Goswami M (2023) Relative estimation of scattering noise and electronic noise of a radiation detector. *Meas Sci Technol* 34(12):125403
- Kyme AZ, Fulton RR (2021) Motion estimation and correction in spect, pet and ct. *Phys Med Biol* 66(18):18TR02

- Kyriakou Y, Meyer E, Prell D, Kachelrieß M (2010) Empirical beam hardening correction (ebhc) for ct. *Med Phys* 37(10):5179–5187
- Law W-Y, Huang G-L, Yang C-C (2022) Effect of body mass index in coronary ct angiography performed on a 256-slice multi-detector ct scanner. *Diagnostics* 12(2):319
- Lee D, Choi S, Kim H-J (2018) Performance evaluation of image denoising developed using convolutional denoising autoencoders in chest radiography. *Nucl Instrum Methods Phys Res, Sect A* 884:97–104
- Lee D, Kim K, Kim W, Kang S, Park C, Cho H, Lim Y, Kim G, Park S, Lim H et al (2019) Four-dimensional cbct reconstruction based on a residual convolutional neural network for improving image quality. *J Korean Phys Soc* 75:73–79
- Lei Y, Niu C, Zhang J, Wang G, Shan H (2023) Ct image denoising and deblurring with deep learning: current status and perspectives. *IEEE Trans Radiat Plasma Med Sci* 8(2):153–172
- Leithner D, Mahmoudi S, Wichmann JL, Martin SS, Lenga L, Albrecht MH, Booz C, Arendt CT, Beeres M, D'Angelo T et al (2018) Evaluation of virtual monoenergetic imaging algorithms for dual-energy carotid and intracerebral ct angiography: effects on image quality, artefacts and diagnostic performance for the detection of stenosis. *Eur J Radiol* 99:111–117
- Lell MM, Kachelrieß M (2020) Recent and upcoming technological developments in computed tomography: high speed, low dose, deep learning, multienergy. *Invest Radiol* 55(1):8–19
- Lell M, Kachelrieß M (2023) Computed tomography 2.0: new detector technology, ai, and other developments. *Invest Radiol* 58(8):587–601
- Li B, Toth TL, Hsieh J, Tang X (2012) Simulation and analysis of image quality impacts from single source, ultra-wide coverage ct scanner. *J Xray Sci Technol* 20(4):395–404
- Li P, Xu L, Yang L, Wang R, Hsieh J, Sun Z, Fan Z, Leipsic JA (2018) Blooming artifact reduction in coronary artery calcification by a new de-blooming algorithm: initial study. *Sci Rep* 8(1):6945
- Li R-F, Hou C-L, Zhou H, Dai Y-S, Jin L-Q, Xi Q, Zhang J-H (2020a) Comparison on radiation effective dose and image quality of right coronary artery on prospective ecg-gated method between 320 row ct and 2nd generation (128-slice) dual source ct. *J Appl Clin Med Phys* 21(8):256–262
- Li M, Hsu W, Xie X, Cong J, Gao W (2020b) Sacnn: self-attention convolutional neural network for low-dose ct denoising with self-supervised perceptual loss network. *IEEE Trans Med Imaging* 39(7):2289–2301
- Li Z, Leng S, Halaweish AF, Yu Z, Yu L, Ritman EL, McCollough CH (2020c) Overcoming calcium blooming and improving the quantification accuracy of percent area luminal stenosis by material decomposition of multi-energy computed tomography datasets. *J Med Imaging* 7(5):053501–053501
- Li D, Ma L, Li J, Qi S, Yao Y, Teng Y (2022d) A comprehensive survey on deep learning techniques in ct image quality improvement. *Med Biol Eng Comput* 60(10):2757–2770
- Li Z, Zhou S, Huang J, Yu L, Jin M (2020e) Investigation of low-dose ct image denoising using unpaired deep learning methods. *IEEE Trans Radiat Plasma Med Sci* 5(2):224–234
- Li M, Du Q, Duan L, Yang X, Zheng J, Jiang H, Li M (2021) Incorporation of residual attention modules into two neural networks for low-dose ct denoising. *Med Phys* 48(6):2973–2990
- Li W, Lu H, Wen Y, Zhou M, Shuai T, You Y, Zhao J, Liao K, Lu C, Li J et al (2023) Reducing both radiation and contrast doses for overweight patients in coronary ct angiography with 80-kvp and deep learning image reconstruction. *Eur J Radiol* 161:110736
- Liang J, Wang H, Xu L, Dong L, Fan Z, Wang R, Sun Z (2018) Impact of ssf on diagnostic performance of coronary computed tomography angiography within 1 heart beat in patients with high heart rate using a 256-row detector computed tomography. *J Comput Assist Tomogr* 42(1):54–61
- Liang J, Sun Y, Ye Z, Sun Y, Xu L, Zhou Z, Thomsen B, Li J, Sun Z, Fan Z (2019) Second-generation motion correction algorithm improves diagnostic accuracy of single-beat coronary ct angiography in patients with increased heart rate. *Eur Radiol* 29:4215–4227
- Lifton J (2017) Multi-material linearization beam hardening correction for computed tomography. *J Xray Sci Technol* 25(4):629–640
- Lin M, Lan Q, Huang C, Yang B, Yu Y (2024) Wavelet-based u-shape network for bioabsorbable vascular stents segmentation in ivoct images. *Front Physiol* 15:1454835
- Liu Y, Zhang Y (2018) Low-dose ct restoration via stacked sparse denoising autoencoders. *Neurocomputing* 284:80–89
- Liu L, Cheng J, Quan Q, Wu F-X, Wang Y-P, Wang J (2020a) A survey on u-shaped networks in medical image segmentations. *Neurocomputing* 409:244–258
- Liu P, Wang M, Wang Y, Yu M, Wang Y, Liu Z, Li Y, Jin Z (2020b) Impact of deep learning-based optimization algorithm on image quality of low-dose coronary ct angiography with noise reduction: a prospective study. *Acad Radiol* 27(9):1241–1248
- Liu X, Xie Y, Cheng J, Diao S, Tan S, Liang X (2023) “Diffusion probabilistic priors for zero-shot low-dose ct image denoising,” Preprint at [arXiv:2305.15887](https://arxiv.org/abs/2305.15887)

- Lossau T, Nickisch H, Wissel T, Bippus R, Schmitt H, Morlock M, Grass M (2019) Motion artifact recognition and quantification in coronary ct angiography using convolutional neural networks. *Med Image Anal* 52:68–79
- Ma Y, Wei B, Feng P, He P, Guo X, Wang G (2020) Low-dose ct image denoising using a generative adversarial network with a hybrid loss function for noise learning. *IEEE Access* 8:67519–67529
- Madesta F, Gauer T, Sentker T, Werner R (2019) “Self-consistent deep learning-based boosting of 4d cone-beam computed tomography reconstruction,” In: *Medical Imaging 2019: Image Processing*, **10949**, p 1094902, SPIE
- Madesta F, Sentker T, Gauer T, Werner R (2020) Self-contained deep learning-based boosting of 4d cone-beam ct reconstruction. *Med Phys* 47(11):5619–5631
- Marcos L, Alirezaie J, Babyn P (2021) “Low dose ct image denoising using boosting attention fusion gan with perceptual loss,” In: *2021 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp 3407–3410, IEEE
- Marcus RP, Nagy DA, Feuerriegel GC, Anhaus J, Nanz D, Sutter R (2024) Photon-counting detector ct with denoising for imaging of the osseous pelvis at low radiation doses: a phantom study. *Am J Roentgenol* 222(1):e2329765
- Marsh Jr J, Rajendran K, Tao S, Vercnocke A, Anderson J, Leng S, Ritman E, McCollough C (2020) “A blooming correction technique for improved vasa vasorum detection using an ultra-high-resolution photon-counting detector ct,” In: *Medical Imaging 2020: Physics of Medical Imaging*, **11312**, pp 1169–1176, SPIE
- Martínez C, Del Cerro CF, Desco M, Abella M (2021) “New method for correcting beam-hardening artifacts in ct images via deep learning,” In: *16th International Meeting on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine*
- Martínez C, Fessler JA, Desco M, Abella M (2022) Simple beam hardening correction method (2dcalbh) based on 2d linearization. *Phys Med Biol* 67(11):115005
- McCollough C (2016) Tu-fg-207a-04: overview of the low dose ct grand challenge. *Med Phys* 43(6Part35):3759–3760
- McCollough CH, Rajendran K, Leng S, Yu L, Fletcher JG, Stierstorfer K, Flohr TG (2023) The technical development of photon-counting detector ct. *Eur Radiol* 33(8):5321–5330
- McLeavy C, Chunara M, Gravell R, Rauf A, Cushnie A, Talbot CS, Hawkins R (2021) The future of ct: deep learning reconstruction. *Clin Radiol* 76(6):407–415
- Meganck JA, Kozloff KM, Thornton MM, Broski SM, Goldstein SA (2009) Beam hardening artifacts in micro-computed tomography scanning can be reduced by x-ray beam filtration and the resulting images can be used to accurately measure bmd. *Bone* 45(6):1104–1116
- Members WC, Mark DB, Berman DS, Budoff MJ, Carr JJ, Gerber TC, Hecht HS, Hlatky MA, Hodgson JM, Lauer MS et al (2010) *Accf/acr/aha/nasci/saip/scai/sct 2010 expert consensus document on coronary computed tomographic angiography: a report of the american college of cardiology foundation task force on expert consensus documents*. *Circulation* 121(22):2509–2543
- Merlin T, Visvikis D, Fernandez P, Lamare F (2015) A novel partial volume effects correction technique integrating deconvolution associated with denoising within an iterative pet image reconstruction. *Med Phys* 42(2):804–819
- Mileto A, Zamora DA, Alessio AM, Pereira C, Liu J, Bhargava P, Carnell J, Cowan SM, Dighe MK, Gunn ML et al (2018) Ct detectability of small low-contrast hypoattenuating focal lesions: iterative reconstructions versus filtered back projection. *Radiology* 289(2):443–454
- Mileto A, Guimaraes LS, McCollough CH, Fletcher JG, Yu L (2019) State of the art in abdominal ct: the limits of iterative reconstruction algorithms. *Radiology* 293(3):491–503
- Min JK, Shaw LJ, Berman DS (2010) The present state of coronary computed tomography angiography: a process in evolution. *J Am Coll Cardiol* 55(10):957–965
- Ming J, Yi B, Zhang Y, Li H (2020) Low-dose ct image denoising using classification densely connected residual network. *KSII Trans Int Inform Syst (TIIS)* 14(6):2480–2496
- Moen TR, Chen B, Holmes DR III, Duan X, Yu Z, Yu L, Leng S, Fletcher JG, McCollough CH (2021) Low-dose ct image and projection dataset. *Med Phys* 48(2):902–911
- Mohammadinejad P, Mileto A, Yu L, Leng S, Guimaraes LS, Missert AD, Jensen CT, Gong H, McCollough CH, Fletcher JG (2021) Ct noise-reduction methods for lower-dose scanning: strengths and weaknesses of iterative reconstruction algorithms and new techniques. *Radiographics* 41(5):1493–1508
- Mori I, Machida Y, Osanai M, Iinuma K (2013) Photon starvation artifacts of x-ray ct: their true cause and a solution. *Radiol Phys Technol* 6:130–141
- Moss AJ, Williams MC, Newby DE, Nicol ED (2017) The updated nice guidelines: cardiac ct as the first-line test for coronary artery disease. *Curr Cardiovasc Imaging Rep* 10:1–7
- Mudeng V, Kim M, Choe S-W (2022) Prospects of structural similarity index for medical image analysis. *Appl Sci* 12(8):3754

- Nagarajappa AK, Dwivedi N, Tiwari R (2015) Artifacts: the downturn of cbct image. *J Int Soc Prevent Commun Dent* 5(6):440–445
- Nagayama Y, Emoto T, Kato Y, Kidoh M, Oda S, Sakabe D, Funama Y, Nakaura T, Hayashi H, Takada S et al (2023) Improving image quality with super-resolution deep-learning-based reconstruction in coronary ct angiography. *Eur Radiol* 33(12):8488–8500
- Nakamura M, Narita Y, Sawada A, Matsugi K, Nakata M, Matsuo Y, Mizowaki T, Hiraoka M (2009) Impact of motion velocity on four-dimensional target volumes: a phantom study. *Med Phys* 36(5):1610–1617
- Nakamura Y, Higaki T, Kondo S, Kawashita I, Takahashi I, Awai K (2023) An introduction to photon-counting detector ct (pcd ct) for radiologists. *Jpn J Radiol* 41(3):266–282
- Nakaura T, Iyama Y, Kidoh M, Yokoyama K, Oda S, Tokuyasu S, Harada K, Yamashita Y (2016) Comparison of iterative model, hybrid iterative, and filtered back projection reconstruction techniques in low-dose brain ct: impact of thin-slice imaging. *Neuroradiology* 58:245–251
- Nannini G, Saitta S, Baggiano A, Maragna R, Mushtaq S, Pontone G, Redaelli A (2024) A fully automated deep learning approach for coronary artery segmentation and comprehensive characterization. *APL Bioeng* 8(1):1
- Nazemi E, Six N, Iuso D, De Samber B, Sijbers J, De Beenhouwer J (2021) Monte-carlo-based estimation of the x-ray energy spectrum for ct artifact reduction. *Appl Sci* 11(7):3145
- Nieman K, García-García HM, Hideo-Kajita A, Collet C, Dey D, Pugliese F, Weissman G, Tijssen JG, Leipsic J, Opolski MP, et al, (2024) “Standards for quantitative assessments by coronary computed tomography angiography (ccta): An expert consensus document of the society of cardiovascular computed tomography (scct),” *J Cardiovasc Comput Tomograph*
- Nikolic B, Khosa F, Lin P-JP, Khan AN, Sarwar S, Yam C-S, Court LE, Raptopoulos V, Clouse ME (2010) Absorbed radiation dose in radiosensitive organs during coronary ct angiography using 320-mdct: effect of maximum tube voltage and heart rate variations. *Am J Roentgenol* 195(6):1347–1354
- Nishii T, Kono AK, Tani W, Suehiro E, Negi N, Takahashi S, Sugimura K (2016) Four-dimensional noise reduction using the time series of medical computed tomography datasets with short interval times: a static-phantom study. *PeerJ* 4:e1680
- Njiti M, Osman N, Mansor M, Rabaiee N, Aziz MA (2024) Potential of metal artifact reduction (mar) and deep learning-based reconstruction (dlr) integration in ct metal artifact correction: A review. *Radiat Phys Chem* 218:111541
- Olasz C, Varga LG, Nagy A (2021) “Beam hardening artifact removal by the fusion of fbp and deep neural networks,” In: Thirteenth International Conference on Digital Image Processing (ICDIP 2021), **11878**, pp 350–360, SPIE
- Oostveen LJ, Meijer FJ, de Lange F, Smit EJ, Pegge SA, Steens SC, van Amerongen MJ, Prokop M, Sechopoulos I (2021) Deep learning-based reconstruction may improve non-contrast cerebral ct imaging compared to other current reconstruction algorithms. *Eur Radiol* 31:5498–5506
- Orii M, Sone M, Osaki T, Ueyama Y, Chiba T, Sasaki T, Yoshioka K (2023) Super-resolution deep learning reconstruction at coronary computed tomography angiography to evaluate the coronary arteries and in-stent lumen: an initial experience. *BMC Med Imaging* 23(1):171
- Otgonbaatar C, Ryu J-K, Kim S, Seo JW, Shim H, Hwang DH (2021) Improvement of depiction of the intracranial arteries on brain ct angiography using deep learning reconstruction. *J Integr Neurosci* 20(4):967–976
- Pack JD, Xu M, Wang G, Baskaran L, Min J, De Man B (2022) Cardiac ct blooming artifacts: clinical significance, root causes and potential solutions. *Visual Comput Indust Biomed Art* 5(1):29
- Padole A, Ali Khawaja RD, Kalra MK, Singh S (2015) Ct radiation dose and iterative reconstruction techniques. *Am J Roentgenol* 204(4):W384–W392
- Pandey RK, Rathore YK (2025) Deep learning in 3d cardiac reconstruction: a systematic review of methodologies and dataset. *Med Biol Eng Comput*. 1–17
- Papadakis AE, Damilakis J (2019) Automatic tube current modulation and tube voltage selection in pediatric computed tomography: a phantom study on radiation dose and image quality. *Invest Radiol* 54(5):265–272
- Park HS, Lee SM, Kim HP, Seo JK, Chung YE (2018) Ct sinogram-consistency learning for metal-induced beam hardening correction. *Med Phys* 45(12):5376–5384
- Park D, Park E-A, Jeong B, Lee YS, Lee W (2024) Quantitative analysis of blooming artifact caused by calcification based on x-ray energy difference using computed tomography. *Sci Rep* 14(1):11539
- Patel N, Li D, Nakanishi R, Fatima B, Andreini D, Pontone G, Conte E, O'Rourke R, Jayawardena E, Hamilton-Craig C et al (2019) Comparison of whole heart computed tomography scanners for image quality lower radiation dosing in coronary computed tomography angiography: the converge registry. *Acad Radiol* 26(11):1443–1449
- Pengpan T, Qiu W, Smith N, Soleimani M (2012) Cone beam ct using motion-compensated algebraic reconstruction methods with limited data. *Comput Methods Programs Biomed* 105(3):246–256

- Pryde MC, Rioux J, Cora AE, Volders D, Schmidt MH, Abdoell M, Bowen C, Beyea SD (2025) Correlation of objective image quality metrics with radiologists' diagnostic confidence depends on the clinical task performed. *J Med Imaging* 12(5):051803–051803
- Rajagopal JR, Farhadi F, Saboury B, Sahbaee P, Negussie AH, Pritchard WF, Jones EC, Samei E (2024) Multivariate signal-to-noise ratio as a metric for characterizing spectral computed tomography. *Phys Med Biol* 69(14):145005
- Rampinelli C, Origgi D, Bellomi M (2012) Low-dose ct: technique, reading methods and image interpretation. *Cancer Imaging* 12(3):548
- Reiter M, de Oliveira FB, Bartscher M, Gusenbauer C, Kastner J (2019) Case study of empirical beam hardening correction methods for dimensional x-ray computed tomography using a dedicated multi-material reference standard. *J Nondestr Eval* 38(1):10
- Ren P, He Y, Zhu Y, Zhang T, Cao J, Wang Z, Yang Z (2022) Motion artefact reduction in coronary ct angiography images with a deep learning method. *BMC Med Imaging* 22(1):184
- Rezaei S, Ghafarian P, Jha AK, Rahmim A, Sarkar S, Ay MR (2019) Joint compensation of motion and partial volume effects by iterative deconvolution incorporating wavelet-based denoising in oncologic pet/ct imaging. *Physica Med* 68:52–60
- Rieke N, Hancox J, Li W, Milletari F, Roth HR, Albarqouni S, Bakas S, Galtier MN, Landman BA, Maier-Hein K et al (2020) The future of digital health with federated learning. *NPJ digital medicine* 3(1):119
- Rinkel J, Gerfault L, Esteve F, Dinten J (2007) A new method for x-ray scatter correction: first assessment on a cone-beam ct experimental setup. *Phys Med Biol* 52(15):4633
- La Riviere PJ (2005) Penalized-likelihood sinogram smoothing for low-dose ct. *Med Phys* 32(6Part1):1676–1683
- Rohkohl C, Bruder H, Stierstorfer K, Flohr T (2013) Improving best-phase image quality in cardiac ct by motion correction with mam optimization. *Med Phys* 40(3):031901
- Sabarudin A, Sun Z, Ng K-H (2013) Coronary computed tomography angiography with prospective electrocardiography triggering: a systematic review of image quality and radiation dose. *Singapore Med J* 54(1):15–23
- Sadia RT, Chen J, Zhang J (2024) Ct image denoising methods for image quality improvement and radiation dose reduction. *J Appl Clin Med Phys* 25(2):e14270
- Saleem AM, Jung S, Chang H-J, Lee S (2024) Cycle consistent generative motion artifact correction in coronary computed tomography angiography. *Appl Sci* 14(5):1859
- Saltybaeva N, Schmidt B, Wimmer A, Flohr T, Alkadhi H (2018) Precise and automatic patient positioning in computed tomography: avatar modeling of the patient surface using a 3-dimensional camera. *Invest Radiol* 53(11):641–646
- Samant S, Panagopoulos AN, Wu W, Zhao S, Chatzizisis YS (2025) Artificial intelligence in coronary artery interventions: preprocedural planning and procedural assistance. *J Soc Cardiovasc Angiograph Interventi* 4(3):102519
- Sandfort V, Persson M, Pourmorteza A, Noël PB, Fleischmann D, Willemlink MJ (2021) Spectral photon-counting ct in cardiovascular imaging. *J Cardiovasc Comput Tomogr* 15(3):218–225
- Sandfort V, Willemlink MJ, Codari M, Mastrodicasa D, Fleischmann D (2024) Denoising multiphase functional cardiac ct angiography using deep learning and synthetic data. *Radiol: Artif Intell* 6(2):230153
- Schroeder S, Achenbach S, Bengel F, Burgstahler C, Cademartiri F, De Feyter P, George R, Kaufmann P, Kopp AF, Knuuti J et al (2008) Cardiac computed tomography: indications, applications, limitations, and training requirements: report of a writing group deployed by the working group nuclear cardiology and cardiac ct of the european society of cardiology and the european council of nuclear cardiology. *Eur Heart J* 29(4):531–556
- Secchi F, De Cecco CN, Spearman JV, Silverman JR, Ebersberger U, Sardanelli F, Schoepf UJ (2015) Monoenergetic extrapolation of cardiac dual energy ct for artifact reduction. *Acta Radiol* 56(4):413–418
- Seeram E (2010) Computed tomography: physical principles and recent technical advances. *J Med Imaging Radiat Sci* 41(2):87–109
- Selles M, van Osch J, Maas M, Boomsma M, Wellenberg R (2023) Advances in metal artifact reduction in ct images: a review of traditional and novel metal artifact reduction techniques. *Eur J Radiol* 170:111276
- Shabestari AA (2013) Coronary artery calcium score: a review. *Iran Red Crescent Med J* 15(12):e16616
- Shameer K, Johnson KW, Glicksberg BS, Dudley JT, Sengupta PP (2018) Machine learning in cardiovascular medicine: are we there yet? *Heart* 104(14):1156–1164
- Shcherbinin S, Celler A (2011) Assessment of the severity of partial volume effects and the performance of two template-based correction methods in a spect/ct phantom experiment. *Phys Med Biol* 56(16):5355
- Shefer E, Altman A, Behling R, Goshen R, Gregorian L, Roterman Y, Uman I, Wainer N, Yagil Y, Zarchin O (2013) State of the art of ct detectors and sources: a literature review. *Curr Radiol Rep* 1(1):76–91

- Sheller MJ, Edwards B, Reina GA, Martin J, Pati S, Kotrotsou A, Milchenko M, Xu W, Marcus D, Colen RR et al (2020) Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. *Sci Rep* 10(1):12598
- Shi D, Zou Y, Zamyatin AA (2011) "Weighted simultaneous algebraic reconstruction technique," In: 11th international meeting on Fully three-dimensional image reconstruction in radiology and nuclear medicine
- Shi Y, Gao Y, Mou X, Liang Z (2020) "Tensor convolutional neural network architecture for spectral ct reconstruction," In: *Medical Imaging 2020: Physics of Medical Imaging*, **11312**, pp 487–492, SPIE
- Shi Y, Gao Y, Xu Q, Li Y, Mou X, Liang Z (2024) Learned tensor neural network texture prior for photon-counting ct reconstruction. *IEEE Trans Med Imaging* 43(11):3830–3842
- Shi S, Kang R, Liatsis P (2025) A conditional diffusion model for electrical impedance tomography image reconstruction. *IEEE Transactions on Instrumentation and Measurement*
- Shrivastava P, Kashikar S, Parihar P, Kasat P, Bhangale P, Shrivastava P (2025) A systematic review on deep learning-enabled coronary ct angiography for plaque and stenosis quantification and cardiac risk prediction. *Europ J Radiol Open* 14:100652
- Singh S, Digumarthy SR, Back A, Shepard J-AO, Kalra MK (2013) Radiation dose reduction for chest ct with non-linear adaptive filters. *Acta Radiol* 54(2):169–174
- Sirajuddin A, Mirmomen SM, Kligerman SJ, Groves DW, Burke AP, Kureshi F, White CS, Arai AE (2021) Ischemic heart disease: noninvasive imaging techniques and findings. *Radiographics* 41(4):990–1021
- Sisniega A, Stayman JW, Yorkston J, Siewerdsen J, Zbijewski W (2017) Motion compensation in extremity cone-beam ct using a penalized image sharpness criterion. *Phys Med Biol* 62(9):3712
- Sisniega A, Capostagno S, Zbijewski W, Stayman J, Weiss C, Ehtiat T, Siewerdsen J (2020) "Estimation of local deformable motion in image-based motion compensation for interventional cone-beam ct," In: *Medical Imaging 2020: Physics of Medical Imaging*, **11312**, pp 385–390, SPIE
- Solomon J, Marin D, Roy Choudhury K, Patel B, Samei E (2017) Effect of radiation dose reduction and reconstruction algorithm on image noise, contrast, resolution, and detectability of subtle hypoattenuating liver lesions at multidetector ct: filtered back projection versus a commercial model-based iterative reconstruction algorithm. *Radiology* 284(3):777–787
- Sone S, Li F, Yang Z, Honda T, Maruyama Y, Takashima S, Hasegawa M, Kawakami S, Kubo K, Haniuda M et al (2001) Results of three-year mass screening programme for lung cancer using mobile low-dose spiral computed tomography scanner. *Br J Cancer* 84(1):25–32
- Spadea MF, Pileggi G, Zaffino P, Salome P, Catana C, Izquierdo-Garcia D, Amato F, Seco J (2019) Deep convolution neural network (dcnn) multiplane approach to synthetic ct generation from mr images-application in brain proton therapy. *Int J Radiat Oncol Biol Phys* 105(3):495–503
- Steckmann S, Kachelrieß M (2010) "Blooming artifact reduction for cardiac ct," In: *IEEE Nuclear Science Symposium & Medical Imaging Conference*, pp 2030–2035, IEEE
- Stehli J, Clerc OF, Fuchs TA, Possner M, Gräni C, Benz DC, Buechel RR, Kaufmann PA (2016) Impact of monochromatic coronary computed tomography angiography from single-source dual-energy ct on coronary stenosis quantification. *J Cardiovasc Comput Tomogr* 10(2):135–140
- Stocker TJ, Leipsic J, Hadamitzky M, Chen MY, Rubinshtein R, Deseive S, Heckner M, Bax JJ, Kitagawa K, Marques H et al (2020) Application of low tube potentials in ccta: results from the protection vi study. *Cardiovasc Imaging* 13:425–434
- Sun Z, Ng CK (2022a) Artificial intelligence (enhanced super-resolution generative adversarial network) for calcium deblooming in coronary computed tomography angiography: a feasibility study. *Diagnostics* 12(4):991
- Sun Z, Ng CK (2022b) Finetuned super-resolution generative adversarial network (artificial intelligence) model for calcium deblooming in coronary computed tomography angiography. *J Personal Med* 12(9):1354
- Sun Z, Choo G, Ng KH (2012) Coronary ct angiography: current status and continuing challenges. *Br J Radiol* 85(1013):495–510
- Sun T, Kim J-H, Fulton R, Nuyts J (2016) An iterative projection-based motion estimation and compensation scheme for head x-ray ct. *Med Phys* 43(10):5705–5716
- Sun T, Jacobs R, Pauwels R, Tijskens E, Fulton R, Nuyts J (2021) A motion correction approach for oral and maxillofacial cone-beam ct imaging. *Phys Med Biol* 66(12):125008
- Taguchi K, Iwanczyk JS (2013) *Vision 20/20: single photon counting x-ray detectors in medical imaging*. *Med Phys* 40(10):100901
- Takafuji M, Kitagawa K, Mizutani S, Hamaguchi A, Kisou R, Iio K, Ichikawa K, Izumi D, Sakuma H (2022) Super-resolution deep learning reconstruction for improved image quality of coronary ct angiography. *Radiol: Cardiothoracic Imaging* 5(4):230085
- Tækker M, Kristjánssdóttir B, Graumann O, Laursen CB, Pietersen PI (2021) Diagnostic accuracy of low-dose and ultra-low-dose ct in detection of chest pathology: a systematic review. *Clin Imaging* 74:139–148

- Tang Q, Cammin J, Srivastava S, Taguchi K (2012) A fully four-dimensional, iterative motion estimation and compensation method for cardiac ct. *Med Phys* 39(7Part1):4291–4305
- Tang C, Li J, Wang L, Li Z, Jiang L, Cai A, Zhang W, Liang N, Li L, Yan B, et al (2019) “Unpaired low-dose ct denoising network based on cycle-consistent generative adversarial network with prior image information,” *Comput Math Methods Med*, 2019
- Tatsugami F, Higaki T, Nakamura Y, Yu Z, Zhou J, Lu Y, Fujioka C, Kitagawa T, Kihara Y, Iida M et al (2019) Deep learning-based image restoration algorithm for coronary ct angiography. *Eur Radiol* 29:5322–5329
- Tesche C, De Cecco CN, Schoepf UJ, Duguay TM, Albrecht MH, De Santis D, Varga-Szemes A, Lesslie VW, Ebersberger U, Bayer RR 2nd et al (2017) Ct coronary calcium scoring with tin filtration using iterative beam-hardening calcium correction reconstruction. *Eur J Radiol* 91:29–34
- Thanh D, Surya P et al (2019) A review on ct and x-ray images denoising methods. *Informatica* 43(2)
- Thomsen M, Knudsen EB, Willendrup PK, Bech M, Willner M, Pfeiffer F, Poulsen M, Lefmann K, Feidenhans R (2015) Prediction of beam hardening artefacts in computed tomography using monte carlo simulations. *Nucl Instrum Methods Phys Res, Sect B* 342:314–320
- Thummerer A, Seller Oria C, Zaffino P, Visser S, Meijers A, Guterres Marmitt G, Wijsman R, Seco J, Langendijk JA, Knopf AC et al (2022) Deep learning–based 4d-synthetic cts from sparse-view cbcts for dose calculations in adaptive proton therapy. *Med Phys* 49(11):6824–6839
- Tootell A, Szczepura K, Hogg P (2014) An overview of measuring and modelling dose and risk from ionising radiation for medical exposures. *Radiography* 20(4):323–332
- Triche BL, Nelson JT Jr, McGill NS, Porter KK, Sanyal R, Tessler FN, McConathy JE, Gauntt DM, Yester MV, Singh SP (2019) Recognizing and minimizing artifacts at ct, mri, us, and molecular imaging. *Radiographics* 39(4):1017–1018
- Usui K, Kamiyama S, Arita A, Ogawa K, Sakamoto H, Sakano Y, Kyogoku S, Daida H (2024) Reducing image artifacts in sparse projection ct using conditional generative adversarial networks. *Sci Rep* 14(1):3917
- Usui K, Ogawa K, Goto M, Sakano Y, Kyougoku S, Daida H (2021) Quantitative evaluation of deep convolutional neural network-based image denoising for low-dose computed tomography. *Visual Comput Indust Biomed Art* 4:1–9
- Van Aarle W, Palenstijn WJ, Cant J, Janssens E, Bleichrodt F, Dabrovolski A, De Beenhouwer J, Batenburg KJ, Sijbers J (2016) Fast and flexible x-ray tomography using the astra toolbox. *Opt Express* 24(22):25129–25147
- Van Herten RLM, van Harten L, Planken N, Isgum I (2024) “Generative adversarial networks for coronary ct angiography acquisition protocol correction with explicit attenuation constraints,” In: *Medical Imaging with Deep Learning*, pp 1288–1303, PMLR
- Van Meter PD, Marsh Jr J, Rajendran K, Leng S, McCollough C (2022) “Quantification of coronary calcification using high-resolution photon-counting-detector ct and an image domain denoising algorithm,” In: *Medical Imaging 2022: Physics of Medical Imaging*, 12031, pp. 433–438, SPIE
- Vavrik D, Jakubek J (2009) Radiogram enhancement and linearization using the beam hardening correction method. *Nucl Instrum Methods Phys Res, Sect A* 607(1):212–214
- Vecsey-Nagy M, Jermendy ÁL, Kolossváry M, Vattay B, Boussousoum M, Suhai FI, Panajotu A, Csöre J, Borzsák S, Fontanini DM et al (2022) Heart rate-dependent degree of motion artifacts in coronary ct angiography acquired by a novel purpose-built cardiac ct scanner. *J Clin Med* 11(15):4336
- Vennart NJ, Bird N, Buscombe J, Cheow HK, Nowosinska E, Heard S (2017) Optimization of pet/ct image quality using the ge ‘sharp ir’ point-spread function reconstruction algorithm. *Nucl Med Commun* 38(6):471–479
- Vey BL, Gichoya JW, Prater A, Hawkins CM (2019) The role of generative adversarial networks in radiation reduction and artifact correction in medical imaging. *J Am Coll Radiol* 16(9):1273–1278
- Wang J, Gu X (2013) Simultaneous motion estimation and image reconstruction (smeir) for 4d cone-beam ct. *Med Phys* 40(10):101912
- Wang J, Li T, Lu H, Liang Z (2006) Penalized weighted least-squares approach to sinogram noise reduction and image reconstruction for low-dose x-ray computed tomography. *IEEE Trans Med Imaging* 25(10):1272–1283
- Wang Y, Li W, Fu S, Zhang C (2015) Adaptive filtering with self-similarity for low-dose ct imaging. *Optik* 126(24):4949–4953
- Wang Z, Vandersteen C, Demarcy T, Gnansia D, Raffaelli C, Guevara N, Delingette H (2019) “Deep learning based metal artifacts reduction in post-operative cochlear implant ct imaging,” In: *Medical Image Computing and Computer Assisted Intervention–MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13–17, 2019, Proceedings, Part VI* 22, pp 121–129, Springer

- Wang C, Sun X, Zhang B, Lai G, Yu D, Su K (2020) "Brain ct image with motion artifact augmentation based on pggan and fbp for artifact detection," In: Advanced Data Mining and Applications: 16th International Conference, ADMA 2020, Foshan, China, November 12–14, 2020, Proceedings 16, pp 370–378, Springer, New York
- Wang Z, Vandersteen C, Demarcy T, Gnansia D, Raffaelli C, Guevara N, Delingette H (2021) Inner-ear augmented metal artifact reduction with simulation-based 3d generative adversarial networks. *Comput Med Imaging Graph* 93:101990
- Wang H, Dalton L, Fan M, Guo R, McClure J, Crandall D, Chen C (2022a) Deep-learning-based workflow for boundary and small target segmentation in digital rock images using unet++ and ik-ebm. *J Petrol Sci Eng* 215:110596
- Wang G, Guo S, Han L, Cekderi AB, Song X, Zhao Z (2022b) Asymptomatic covid-19 ct image denoising method based on wavelet transform combined with improved pso. *Biomed Signal Process Control* 76:103707
- Wang H, Shang Y, Yuan Z, Wu J, Yan J, Yan Y (2024a) "Quest: Low-bit diffusion model quantization via efficient selective finetuning," Preprint at [arXiv:2402.03666](https://arxiv.org/abs/2402.03666)
- Wang Y, Yoon S, Jin P, Tivnan M, Chen Z, Hu R, Zhang L, Chen Z, Li Q, Wu D (2024b) "Implicit image-to-image schrodinger bridge for ct super-resolution and denoising," Preprint at [arXiv:2403.06069](https://arxiv.org/abs/2403.06069)
- Watkins WT, Li R, Lewis J, Park JC, Sandhu A, Jiang SB, Song WY (2010) Patient-specific motion artifacts in 4dct. *Med Phys* 37(6Part1):2855–2861
- Weir-McCall JR, Wang R, Halankar J, Hsieh J, Hague CJ, Rosenblatt S, Fan Z, Sellers SL, Murphy DT, Blanke P et al (2020) Effect of a calcium deblooming algorithm on accuracy of coronary computed tomography angiography. *J Cardiovasc Comput Tomogr* 14(2):131–136
- Weis M, Henzler T, Nance JW Jr, Haubenreisser H, Meyer M, Sudarski S, Schoenberg SO, Neff KW, Hagelstein C (2017) Radiation dose comparison between 70 kvp and 100 kvp with spectral beam shaping for non-contrast-enhanced pediatric chest computed tomography: a prospective randomized controlled study. *Invest Radiol* 52(3):155–162
- Wen B, Xu L, Liang J, Fan Z, Sun Z (2018) A preliminary study of computed tomography coronary angiography within a single cardiac cycle in patients with atrial fibrillation using 256-row detector computed tomography. *J Comput Assist Tomogr* 42(2):277–281
- Wuest W, May MS, Scharf M, Layritz C, Eisentopf J, Ropers D, Pflederer T, Uder M, Achenbach S, Lell MM (2013) Stent evaluation in low-dose coronary ct angiography: effect of different iterative reconstruction settings. *J Cardiovasc Comput Tomogr* 7(5):319–325
- Xie Q, Zeng D, Zhao Q, Meng D, Xu Z, Liang Z, Ma J (2017) Robust low-dose ct sinogram preprocessing via exploiting noise-generating mechanism. *IEEE Trans Med Imaging* 36(12):2487–2498
- Xie X, Zhang L, Chen Q, Jiang B (2020) Preliminary study on motion artifacts removal of coronary artery cta using generative adversarial network. *J Cardiovasc Comput Tomogr* 14(3):S19
- Xia W, Lyu Q, Wang G (2022) "Low-dose ct using denoising diffusion probabilistic model for 20x speedup," Preprint at [arXiv:2209.15136](https://arxiv.org/abs/2209.15136)
- Xu Q, Xie K, Yuan K, Yu L, Wang W, Ye D (2012) A statistical study of the factors influencing the extent of respiratory motion blur in pet imaging. *Comput Biol Med* 42(1):8–18
- Xu BL, Zhou WL, Zhu TP, Cheng KY, Li YJ, Zhan HJ, Jiang LG, Tong YH (2019) A full-width half-maximum method to assess retinal vascular structural changes in patients with ischemic heart disease and microvascular angina. *Sci Rep* 9(1):11019
- Xu L, Xie J, Cai F, Wu J (2021) Spectral classification based on deep learning algorithms. *Electronics* 10(16):1892
- Yamaguchi S, Ichikawa Y, Takafuji M, Sakuma H, Kitagawa K (2024) Usefulness of second-generation motion correction algorithm in improving delineation and reducing motion artifact of coronary computed tomography angiography. *J Cardiovasc Comput Tomogr* 18(3):281–290
- Yan RT, Miller JM, Rochitte CE, Dewey M, Niinuma H, Clouse ME, Vavere AL, Brinker J, Lima JA, Arbab-Zadeh A (2013) Clinical and imaging characteristics associated with discordance between ct angiography and quantitative coronary angiography on the diagnosis of significant coronary artery disease. *JACC Cardiovasc Imaging* 6(9):963
- Yang DH (2021) Application of artificial intelligence to cardiovascular computed tomography. *Korean J Radiol* 22(10):1597
- Yang L, Shangguan H, Zhang X, Wang A, Han Z (2019) High-frequency sensitive generative adversarial network for low-dose ct image denoising. *IEEE Access* 8:930–943
- Yang D, Lv W, Zhang J, Chen H, Sun X, Lv S, Dai X, Luo R, Zhou W, Qiu J et al (2023) Low-dose imaging denoising with one pair of noisy images. *Opt Express* 31(9):14159–14173
- Yi X, Babyn P (2018) Sharpness-aware low-dose ct denoising using conditional generative adversarial network. *J Digit Imaging* 31:655–669

- Yi Y, Xu C, Xu M, Yan J, Li Y-Y, Wang J, Yang S-J, Guo Y-B, Wang Y, Li Y-M et al (2021) Diagnostic improvements of deep learning-based image reconstruction for assessing calcification-related obstructive coronary artery disease. *Front Cardiovasc Med* 8:758793
- Yin X, Zhao Q, Liu J, Yang W, Yang J, Quan G, Chen Y, Shu H, Luo L, Coatrieux J-L (2019) Domain progressive 3d residual convolution network to improve low-dose ct imaging. *IEEE Trans Med Imaging* 38(12):2903–2913
- You C, Yang Q, Shan H, Gjestebj L, Li G, Ju S, Zhang Z, Zhao Z, Zhang Y, Cong W et al (2018) Structurally-sensitive multi-scale deep neural network for low-dose ct denoising. *IEEE Access* 6:41839–41855
- You C, Zhao R, Liu F, Dong S, Chinchali S, Topcu U, Staib L, Duncan J (2022a) Class-aware adversarial transformers for medical image segmentation. *Adv Neural Inf Process Syst* 35:29582–29596
- You C, Zhao R, Staib LH, Duncan JS (2022b) “Momentum contrastive voxel-wise representation learning for semi-supervised volumetric medical image segmentation,” In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pp. 639–652, Springer
- You C, Dai W, Liu F, Min Y, Dvornek NC, Li X, Clifton DA, Staib L, Duncan JS (2024) “Mine your own anatomy: Revisiting medical image segmentation with extremely limited labels,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*
- Yu H, Wang G (2007) Data consistency based rigid motion artifact reduction in fan-beam ct. *IEEE Trans Med Imaging* 26(2):249–260
- Zavala-Mondragon LA, Rongen P, Bescos JO, de With PH, van der Sommen F (2022) Noise reduction in ct using learned wavelet-frame shrinkage networks. *IEEE Trans Med Imaging* 41(8):2048–2066
- Zeng GL (2022a) A projection-domain iterative algorithm for metal artifact reduction by minimizing the total-variation norm and the negative-pixel energy. *Visual Comput Indus Biomed Art* 5(1):1
- Zeng GL (2022b) Photon starvation artifact reduction by shift-variant processing. *IEEE Access* 10:13633–13649
- Zhang Y, Yu H (2018) Convolutional neural network based metal artifact reduction in x-ray computed tomography. *IEEE Trans Med Imaging* 37(6):1370–1381
- Zhang Y, Yang J, Zhang L, Court LE, Balter PA, Dong L (2013) Modeling respiratory motion for reducing motion artifacts in 4d ct images. *Med Phys* 40(4):041716
- Zhang H, Zhang L, Sun Y, Zhang J (2015) Projection domain denoising method based on dictionary learning for low-dose ct image reconstruction. *J Xray Sci Technol* 23(5):567–578
- Zhang C, Xing Y (2018) “Ct artifact reduction via u-net cnn,” In: *Medical Imaging 2018: Image Processing*, 10574, pp 440–445, SPIE
- Zhao T, Hoffman J, McNitt-Gray M, Ruan D (2019a) Ultra-low-dose ct image denoising using modified bm3d scheme tailored to data statistics. *Med Phys* 46(1):190–198
- Zhang Y, Huang X, Wang J (2019b) Advanced 4-dimensional cone-beam computed tomography reconstruction by combining motion estimation, motion-compensated reconstruction, biomechanical modeling and deep learning. *Visual Comput Indus Biomed Art* 2(1):23
- Zhang Z, Huang M, Jiang Z, Chang Y, Lu K, Yin F-F, Tran P, Wu D, Beltran C, Ren L (2022) Patient-specific deep learning model to enhance 4d-cbct image for radiomics analysis. *Phys Med Biol* 67(8):085003
- Zhang L, Jiang B, Chen Q, Wang L, Zhao K, Zhang Y, Vliegthart R, Xie X (2023a) Motion artifact removal in coronary ct angiography based on generative adversarial networks. *Eur Radiol* 33(1):43–53
- Zhang Z, Liu J, Yang D, Kamilov US, Hugo GD (2023b) Deep learning-based motion compensation for four-dimensional cone-beam computed tomography (4d-cbct) reconstruction. *Med Phys* 50(2):808–820
- Zhi S, Duan J, Cai J, Mou X (2019) “Artifacts reduction method for phase-resolved cone-beam ct (cbct) images via a prior-guided cnn,” In: *Medical Imaging 2019: Physics of Medical Imaging*, 10948, pp 562–567, SPIE
- Zhi S, Mou X (2021) “tn-net: a spatiotemporal plus prior image-based convolutional neural network for 4d-cbct reconstructions enhancement,” In: *Medical Imaging 2021: Physics of Medical Imaging*, 11595, pp 474–481, SPIE
- Zhi S, Kachelrieß M, Pan F, Mou X (2021a) Cycin-net: A convolutional neural network specialized for 4d cbct images refinement. *IEEE Trans Med Imaging* 40(11):3054–3064
- Zhou S, Chi Y, Wang J, Jin M (2021b) General simultaneous motion estimation and image reconstruction (g-smeir). *Biomed Phys Eng Express* 7(5):055011
- Zhu Y, Zhao M, Zhao Y, Li H, Zhang P (2012) Noise reduction with low dose ct data based on a modified rof model. *Opt Express* 20(16):17987–18004
- Ziabari A, Ye DH, Srivastava S, Sauer KD, Thibault J-B, Bouman CA (2018) “2.5 d deep learning for ct image reconstruction using a multi-gpu implementation,” In: *2018 52nd Asilomar Conference on Signals, Systems, and Computers*, pp 2044–2049, IEEE

- Ziabari A, Venkatakrishnan S, Kirka M, Brackman P, Dehoff R, Bingham P, Paquit V (2020) “Beam hardening artifact reduction in x-ray ct reconstruction of 3d printed metal parts leveraging deep learning and cad models,” In: ASME International Mechanical Engineering Congress and Exposition, **84492**, p V02BT02A043, American Society of Mechanical Engineers
- Zijlstra E, Hagedoorn M, Krijnen WP, van der Schans CP, Mobach MP (2017) Motion nature projection reduces patient’s psycho-physiological anxiety during ct imaging. *J Environ Psychol* 53:168–176

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH ("Springer Nature").

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users ("Users"), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use ("Terms"). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval, sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com